

	A	B	C	D	E	F
1	Olympic Bikes Inc.					
2		LHS Coefficients				
3	Constraints	D	P	LHS Value		RHS Value
4	Aluminum	2	4	100	<=	100
5	Steel	3	2	80	<=	80
6	Obj Func Coeff.	10	15			
7						
8		Optimal Decision Variables				
9		D	P			
10		15	17.5			
11						
12	Objective Value		412.5			
13						

Example 2: Olympic Bike Co.

Olympic Bike is introducing two new lightweight bicycle frames, the Deluxe and the Professional, to be made from special aluminum and steel alloys. The anticipated unit profits are \$10 for the Deluxe and \$15 for the Professional. A supplier delivers 100 pounds of the aluminum alloy and 80 pounds of the steel alloy weekly.

	Aluminum Alloy	Steel Alloy
Deluxe	2	3
Professional	4	2



How many Deluxe and Professional frames should Olympic produce each week?

Range of Optimality



Question:

If the unit profit on deluxe frames were \$6 instead of \$10, would the optimal solution change?

Answer:

The output states that the solution remains optimal as long as the objective function coefficient of x_1 is between 7.5 and 22.5. Because 6 is outside this range, the optimal solution may change.

Variable Cells

Cell	Name	Final Value	Reduced Cost	Objective Coefficient	Allowable Increase	Allowable Decrease
\$B\$10 D		15	0	10	12.5	2.5
\$C\$10 P		17.5	0	15	5	8.333333333

Constraints

Cell	Name	Final Value	Shadow Price	Constraint R.H. Side	Allowable Increase	Allowable Decrease
\$D\$4	Aluminum LHS Value	100	3.125	100	60	46.66666667
\$D\$5	Steel LHS Value	80	1.25	80	70	30

Range of Optimality



Question:

Suppose the profit on deluxe frames is increased to \$20. Is the above solution still optimal? What is the value of the objective function when this unit profit is increased to \$20?

Answer:

The output states that the solution remains optimal as long as the objective function coefficient of D

is between 7.5 and 22.5. Because 20 is within this range, the optimal solution will not change. The optimal profit will change: $20D + 15P = 20(15) + 15(17.5) = \562.50 .

Question: If simultaneously the profit on Deluxe frames was raised to \$16 and the profit on Professional frames was raised to \$17, would the current solution be optimal?



Answer:

If $c_1 = 16$, the amount c_1 changed is $16 - 10 = 6$. The maximum allowable increase is $22.5 - 10 = 12.5$, so this is a $6/12.5 = 48\%$ change. If $c_2 = 17$, the amount that c_2 changed is $17 - 15 = 2$. The maximum allowable increase is $20 - 15 = 5$ so this is a $2/5 = 40\%$ change.

The sum of the change percentages is 88%. Since this does not exceed 100%, the optimal solution would not change.

Adjustable Cells					
\$D\$5	Steel	LHS Value	80	1.25	80
			70		30

Example 3: M&D Chemicals

Range of Optimality and 100% Rule

Question:

If simultaneously the cost of A was raised to \$2.5 and the cost of B was reduced to \$2.6, would the current solution remain optimal?

Answer:

- Since $c_1 = 2$, the amount c_1 changed is 0.5. The maximum allowable increase is $3 - 2 = 1$, so this is a $0.5/1 = 50\%$ change

- Since $c_2 = 3$, the amount that c_2 changed is -0.4. The maximum allowable decrease is $3 - 2 = 1$, so this is a $0.4/1 = 40\%$ change.
- The sum of the change percentages is $50\% + 40\% = 90\%$.
- Because this does not exceed 100%, the optimal solution would not change.

Variable Cells

Cell	Name	Final Value	Reduced Cost	Objective Coefficient	Allowable Increase	Allowable Decrease
\$B\$11 A		250	0	2	1	1E+30
\$C\$11 B		100	0	3	1E+30	1

Constraints

Cell	Name	Final Value	Shadow Price	Constraint R.H. Side	Allowable Increase	Allowable Decrease
\$D\$4	Demand for product A LHS Value	250	0	125	125	1E+30
\$D\$5	Total production LHS Value	350	4	350	125	50
\$D\$6	Processing time LHS Value	600	-1	600	100	125

M&D Chemicals					
	LHS Coefficients				
Constraints	A	B	LHS Value		RHS Value
Demand for product A	1	0	250	>=	125
Total production	1	1	350	>=	350
Processing time	2	1	600	<=	600
Obj Func Coeff	2	3			
	Decision Variables				
	A	B			
	250	100			
Minimized Obj Value		800			

• Range of Feasibility and 100% Rule

Question:

If the right-hand side of constraint 2 is decreased by 25, and the right-hand side of constraint 3 is increased by 30, what will be the effect on the optimal solution?

Answer:

- Constraint 2: allowable decrease is 50, a decrease of 25 is within the range, causing a 50% change
- Constraint 3: allowable increase is 100, an increase of 30 is within the range, causing a 30% change
- Since the total change is 80%, the shadow prices are valid. While the new total cost is

$$800 - 25 \cdot 4 - 30 \cdot 1 = \$670$$

Winslow: Portfolio Selection



Winslow Savings has \$20 million available for investment. It wishes to invest over the next four months in such a way that it will maximize the total accrued interests earned over the four month period as well as have at least \$10 million available at the start of the fifth month for a high rise building venture in which it will be participating.

For the time being, Winslow wishes to invest only in 2-month government bonds (earning 2% over the 2-month period) and 3-month construction loans (earning 6% over the 3-month period). Each of these is available each month for investment. Funds not invested in these two investments are liquid and earn $3/4$ of 1% per month when invested locally.

Formulate a linear program that will help Winslow Savings determine how to invest over the next four months if at no time does it wish to have more than \$8 million in either government bonds or construction loans.



- Define the Decision Variables

G_i = amount of new investment in government bonds in month i (for $i = 1, 2, 3, 4$)

C_i = amount of new investment in construction loans in month i (for $i = 1, 2, 3, 4$)

L_i = amount invested locally in month i , (for $i = 1, 2, 3, 4$)

- Define the Objective Function

Maximize total interest earned in the 4-month period:

Max (interest rate on investment) X (amount invested)

$$\begin{aligned} \text{Max} \quad & 0.02G_1 + 0.02G_2 + 0.02G_3 + 0.02G_4 \\ & + 0.06C_1 + 0.06C_2 + 0.06C_3 + 0.06C_4 \\ & + 0.0075L_1 + 0.0075L_2 + 0.0075L_3 + 0.0075L_4 \end{aligned}$$

- Define the Constraints (continued)

Month 4's total investment limited to principle and interest invested in construction loans in Month 1, government bonds in Month 2, and locally invested in Month 3:

$$(4) \quad G_4 + C_4 + L_4 = 1.06C_1 + 1.02G_2 + 1.0075L_3$$

or $- 1.02G_2 + G_4 - 1.06C_1 + C_4 - 1.0075L_3 + L_4 = 0$

\$10 million must be available at start of Month 5:

$$(5) \quad 1.06C_2 + 1.02G_3 + 1.0075L_4 \geq 10,000,000$$

- Define the Constraints (continued)

Month 1's total investment limited to \$20 million:

$$(1) \quad G_1 + C_1 + L_1 = 20,000,000$$

Month 2's total investment limited to principle and interest invested locally in Month 1:

$$(2) \quad G_2 + C_2 + L_2 = 1.0075L_1$$

or $G_2 + C_2 - 1.0075L_1 + L_2 = 0$

Month 3's total investment amount limited to principle and interest invested in government bonds in Month 1 and locally invested in Month 2:

$$(3) \quad G_3 + C_3 + L_3 = 1.02G_1 + 1.0075L_2$$

or $- 1.02G_1 + G_3 + C_3 - 1.0075L_2 + L_3 = 0$

No more than \$8 million in government bonds

$$(6) \quad G_1 \leq 8,000,000$$

$$(7) \quad G_1 + G_2 \leq 8,000,000$$

$$(8) \quad G_2 + G_3 \leq 8,000,000$$

$$(9) \quad G_3 + G_4 \leq 8,000,000$$

No more than \$8 million in construction loans

$$(10) \quad C_1 \leq 8,000,000$$

$$(11) \quad C_1 + C_2 \leq 8,000,000$$

$$(12) \quad C_1 + C_2 + C_3 \leq 8,000,000$$

$$(13) \quad C_2 + C_3 + C_4 \leq 8,000,000$$

Non-negativity:

$$G_i, C_i, L_i \geq 0 \text{ for } i = 1, 2, 3, 4$$

LACET SOLUTION

[illegible]

Cell	Name	Final Value	Reduced Cost	Objective Coefficient	Allowable Increase	Allowable Decrease
\$B\$20 G1		8000000	0	0.02	1E+30	3.7635E-05
\$C\$20 G2		0	0	0.02	3.7635E-05	0.005018002
\$D\$20 G3		5108613.923	0	0.02	0.00755625	0.005005087
\$E\$20 G4		2891386.077	0	0.02	0.007142716	0.00755625
\$F\$20 C1		8000000	0	0.06	1E+30	0.037891513
\$G\$20 C2		0	-0.007371205	0.06	0.007371205	1E+30
\$H\$20 C3		0	-0.007612645	0.06	0.007612645	1E+30
\$I\$20 C4		8000000	0	0.06	1E+30	0.007371205
\$J\$20 L1		4000000	0	0.0075	3.7635E-05	1E+30
\$K\$20 L2		4030000	0	0.0075	0.005018002	0.007371205
\$L\$20 L3		7111611.077	0	0.0075	0.005005087	3.71711E-05
\$M\$20 L4		4753562.083	0	0.0075	0.00494375	1.0075

Constraints						
Cell	Name	Final Value	Shadow Price	Constraint R.H. Side	Allowable Increase	Allowable Decrease
\$N\$4	Total Investment - Month 1 LHS	20000000	0.038029001	20000000	5019884.99	2841167.054
\$N\$5	Total Investment - Month 2 LHS	-4.65661E-10	0.030301738	0	5057534.127	2862475.807
\$N\$6	Total Investment - Month 3 LHS	0	0.022631998	0	5095465.633	2883944.375
\$N\$7	Total Investment - Month 4 LHS	-9.31323E-10	0.015019353	0	5133681.625	2905573.958
\$N\$8	Month 5 Requirement LHS	10000000	-0.007463378	10000000	2927365.763	5172184.237
\$N\$9	Max G-Bond-Month 1 LHS	8000000	3.7635E-05	8000000	0	2891279.397
\$N\$10	Max G-Bond-Month 2 LHS	8000000	0.005018002	8000000	2905680.893	0
\$N\$11	Max G-Bond-Month 3 LHS	5108613.923	0	8000000	1E+30	2891386.077
\$N\$12	Max G-Bond-Month 4 LHS	8000000	0.004980647	8000000	2905573.958	5133681.625
\$N\$13	Max C-Loans-Month 1 LHS	8000000	0.037891513	8000000	0	8000000
\$N\$14	Max C-Loans-Month 2 LHS	8000000	0	8000000	1E+30	0
\$N\$15	Max C-Loans-Month 3 LHS	8000000	0	8000000	1E+30	0
\$N\$16	Max C-Loans-Month 4 LHS	8000000	0.044980647	8000000	2905573.958	5133681.625

NWC: Workforce Assignment

National Wing Company (NWC) is gearing up for the new B-48 contract. NWC has agreed to produce 20 wings in April, 24 in May, and 30 in June. Currently, NWC has 100 fully qualified workers. A fully qualified worker can either be placed in production or can train new recruits. A new recruit can be trained to be an apprentice in one month. After another month, the apprentice becomes a qualified worker. Each trainer can train two recruits.



The production rate and salary per employee type is listed below:

<u>Type of Employee</u>	<u>Production Rate (Wings/Month)</u>	<u>Wage Per Month</u>
Production	0.60	\$3,000
Trainer	0.30	\$3,300
Apprentice	0.40	\$2,600
Recruit	0.05	\$2,200

At the end of June, NWC wishes to have no recruits, but have at least 140 full-time workers.

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Define the Constraints

Total production in Month 1 (April) must equal or exceed contract for Month 1:

$$(1) \quad 0.6P_1 + 0.3T_1 + 0.05R_1 \geq 20$$

NWC: Workforce Assignment

Define the Decision Variables

P_i = number of producers in month i
(where $i = 1, 2, 3$ for April, May, June)

T_i = number of trainers in month i
(where $i = 1, 2$ for April, May)

A_i = number of apprentices in month i
(where $i = 2, 3$ for May, June)

R_i = number of recruits in month i
(where $i = 1, 2$ for April, May)

Total production in Months 1-2 (April, May) must equal or exceed total contracts for Months 1-2:

$$(2) \quad 0.6P_1 + 0.3T_1 + 0.05R_1 + 0.6P_2 + 0.3T_2 + 0.4A_2 + 0.05R_2 \geq 44$$

Total production in Months 1-3 (April, May, June) must equal or exceed total contracts for Months 1-3:

$$(3) \quad 0.6P_1 + 0.3T_1 + 0.05R_1 + 0.6P_2 + 0.3T_2 + 0.4A_2 + 0.05R_2 + 0.6P_3 + 0.4A_3 \geq 74$$

Define the Objective Function

Minimize total wage cost for producers, trainers, apprentices, and recruits for April, May, and June:

$$\begin{aligned} \text{Min} \quad & 3000P_1 + 3300T_1 + 2200R_1 + 3000P_2 + 3300T_2 \\ & + 2600A_2 + 2200R_2 + 3000P_3 + 2600A_3 \end{aligned}$$

Define the Constraints (continued)

The number of producers and trainers in a month must equal the number of producers, trainers, and apprentices in the previous month:

$$(4) P_1 - P_2 + T_1 - T_2 = 0$$

$$(5) P_2 - P_3 + T_2 + A_2 = 0$$

The number of apprentices in a month must equal the number of recruits in the previous month:

$$(6) A_2 - R_1 = 0$$

$$(7) A_3 - R_2 = 0$$

Each trainer can train two recruits:

$$(8) 2T_1 - R_1 \geq 0$$

$$(9) 2T_2 - R_2 \geq 0$$

In April there are 100 employees that can be producers or trainers:

$$(10) P_1 + T_1 = 100$$

At the end of June, there are to be at least 140 employees:

$$(11) P_3 + A_3 \geq 140$$

Non-negativity:

$$P_1, T_1, R_1, P_2, T_2, A_2, R_2, P_3, A_3 \geq 0$$

NWC: Workforce Assignment

	A	B	C	D	E	F	G	H	I	J	K	L	M
1	Lecture 4: NWC (Workforce Assignment)							Excel Solution					
2													
3	Constraints	P1	T1	R1	P2	T2	A2	R2	P3	A3	LHS		
4	Total Production - April	0.6	0.3	0.05							60	>=	20
5	Total Production - April & May	0.6	0.3	0.05	0.6	0.3	0.4	0.05			116	>=	44
6	Total Production - April, May & June	0.6	0.3	0.05	0.6	0.3	0.4	0.05	0.6	0.4	192	>=	74
7	Balance-April to May	1	1		-1	-1					-2.8E-14	=	0
8	Balance-May to June				1	1	1		-1		1.42E-14	=	0
9	A-R Balance - April to May			-1			1				0	=	0
10	A-R Balance - May to June							-1		1	0	=	0
11	T-R Ratio - April		2	-1							0	>=	0
12	T-R Ratio - May					2		-1			0	>=	0
13	Total Number of Employees	1	1								100	=	100
14	Min Number of Employees-June								1	1	140	>=	140
15	Cost	3000	3300	2200	3000	3300	2600	2200	3000	2600			
16													
17	Decision variables	P1	T1	R1	P2	T2	A2	R2	P3	A3			
18		100	0	0	80	20	0	40	100	40			
19													
20	Objective value (min)		1098000										

Solution Summary

$$P_1 = 100, T_1 = 0, R_1 = 0$$

$$P_2 = 80, T_2 = 20, A_2 = 0, R_2 = 40$$

$$P_3 = 100, A_3 = 40$$

$$\text{Total Wage Cost} = \$1,098,000$$

	Apr	May	Jun	Jul
Producers	100	80	100	140
Trainers	0	20	0	0
Apprentices	0	0	40	0
Recruits	0	40	0	0

Example: Tina's Tailoring

Tina's Tailoring has five idle tailors and four custom garments to make. The estimated time (in hours) it would take each tailor to make each garment is shown in the next slide. (An 'X' in the table indicates an unacceptable tailor-garment assignment.)

<u>Garment</u>	<u>Tailor</u>				
	<u>1</u>	<u>2</u>	<u>3</u>	<u>4</u>	<u>5</u>
Wedding gown	19	23	20	21	18
Clown costume	11	14	X	12	10
Admiral's uniform	12	8	11	X	9
Bullfighter's outfit	X	20	20	18	21



Formulate an integer program for determining the tailor-garment assignments that minimize the total estimated time spent making the four garments. No tailor is to be assigned more than one garment and each garment is to be worked on by only one tailor.

Example: Tina's Tailoring

This problem can be formulated as a 0-1 integer program. The LP solution to this problem will automatically be integer (0-1).



- Define the decision variables

$x_{ij} = 1$ if garment i is assigned to tailor j
 $= 0$ otherwise.

Number of decision variables =

$$[(\text{number of garments})(\text{number of tailors})] \\ - (\text{number of unacceptable assignments}) = [4(5)] - 3 = 17$$

- Define the objective function

Minimize total time spent making garments:

$$\begin{aligned} \text{Min } & 19x_{11} + 23x_{12} + 20x_{13} + 21x_{14} + 18x_{15} + 11x_{21} \\ & + 14x_{22} + 12x_{24} + 10x_{25} + 12x_{31} + 8x_{32} + 11x_{33} \\ & + 9x_{35} + 20x_{42} + 20x_{43} + 18x_{44} + 21x_{45} \end{aligned}$$

Define the Constraints

Exactly one tailor per garment:

- 1) $x_{11} + x_{12} + x_{13} + x_{14} + x_{15} = 1$
- 2) $x_{21} + x_{22} + x_{24} + x_{25} = 1$
- 3) $x_{31} + x_{32} + x_{33} + x_{35} = 1$
- 4) $x_{42} + x_{43} + x_{44} + x_{45} = 1$

No more than one garment per tailor

- 5) $x_{11} + x_{21} + x_{31} \leq 1$
- 6) $x_{12} + x_{22} + x_{32} + x_{42} \leq 1$
- 7) $x_{13} + x_{33} + x_{43} \leq 1$
- 8) $x_{14} + x_{24} + x_{44} \leq 1$
- 9) $x_{15} + x_{25} + x_{35} + x_{45} \leq 1$

Non-negativity: $x_{ij} \geq 0$ for $i = 1, \dots, 4$ and $j = 1, \dots, 5$

a

Garment	Tailor 1	Tailor 2	Tailor 2	Tailor 4	Tailor 5
Wedding gown	19	23	20	21	18
Clown costume	11	14	1000	12	10
Admiral's uniform	12	8	11	1000	9
Bullfigher's outfit	1000	20	20	18	21
Model					
Total Servcie Time (min)		55			
Garment	Tailor 1	Tailor 2	Tailor 2	Tailor 4	Tailor 5
Wedding gown	1	0	0	0	0
Clown costume	0	0	0	0	1
Admiral's uniform	0	1	0	0	0
Bullfigher's outfit	0	0	0	1	0
	1	1	0	1	1
	<=	<=	<=	<=	<=
	1	1	1	1	1

Solve

Set Objective:

To: ☐ Max ☒ Min ☐

By Changing Variable Cells:

Subject to the Constraints:

\$B\$15:\$F\$18 = binary

\$B\$19:\$F\$19 <= \$B\$21:\$F\$21

\$G\$15:\$G\$18 = \$I\$15:\$I\$18

1	=	1
1	=	1
1	=	1
1	=	1

Modeling Flexibility Provided by 0-1 Variables

- When x_i and x_j represent binary variables designating whether projects i and j have been completed, the following special constraints may be formulated:
 - At most k out of n projects will be completed:

$$\sum_{j=1}^n x_j \leq k$$

- Project j is conditional on project i :

$$x_j - x_i \leq 0$$

- Project i is a corequisite for project j :

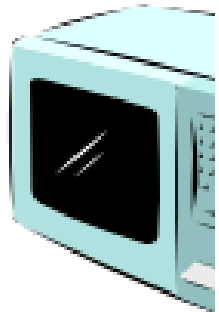
$$x_j - x_i = 0$$

- Projects i and j are mutually exclusive:

$$x_j + x_i \leq 1$$

Example: Metropolitan Microwaves

Metropolitan Microwaves, Inc. is planning to expand its sales operation by offering other electronic appliances. The company has identified seven new product lines it can carry. Relevant information about each line follows on the next slide.



Product Line	Initial Invest.	Floor Space (Sq.Ft.)	Exp. Rate of Return
1. TV/VCRs	\$ 6,000	125	8.1%
2. TVs	12,000	150	9.0
3. Projection TVs	20,000	200	11.0
4. VCRs	14,000	40	10.2
5. DVD Players	15,000	40	10.5
6. Video Games	2,000	20	14.1
7. Home Computers	32,000	100	13.2

^a Example: Metropolitan Microwaves

Metropolitan has decided that they should not stock projection TVs unless they stock either TV/VCRs or TVs. Also, they will not stock both VCRs and DVD players, and they will consider stocking video games only if they stock TVs. Finally, the company wishes to introduce at least three new product lines.

If the company has \$45,000 to invest and 420 sq. ft. of floor space available, formulate an integer linear program for Metropolitan to maximize its overall expected return.

Decision Variables

$x_j = 1$ if product line j is introduced;
= 0 otherwise.

where:

Product line 1 = TV/VCRs

Product line 2 = TVs

Product line 3 = Projection TVs

Product line 4 = VCRs

Product line 5 = DVD Players

Product line 6 = Video Games

Product line 7 = Home Computers

2

□ Define the Objective Function



Maximize total expected return (in '000):

$$\begin{aligned} \text{Max } & 0.081(6)x_1 + 0.09(12)x_2 + 0.11(20)x_3 + 0.102(14)x_4 \\ & + 0.105(15)x_5 + 0.141(2)x_6 + 0.132(32)x_7 \end{aligned}$$

Define the Constraints

1) Money:

$$6x_1 + 12x_2 + 20x_3 + 14x_4 + 15x_5 + 2x_6 + 32x_7 \leq 45$$

2) Space:

$$125x_1 + 150x_2 + 200x_3 + 40x_4 + 40x_5 + 20x_6 + 100x_7 \leq 420$$

3) Stock projection TVs only if stock TV/VCRs or TVs:

$$x_1 + x_2 \geq x_3 \text{ or } x_1 + x_2 - x_3 \geq 0$$

Define the Constraints (continued)



4) Do not stock both VCRs and DVD players:

$$x_4 + x_5 \leq 1$$

5) Stock video games if they stock TV's:

$$x_2 - x_6 \geq 0$$

6) Introduce at least 3 new lines:

$$x_1 + x_2 + x_3 + x_4 + x_5 + x_6 + x_7 \geq 3$$

7) Variables are 0 or 1:

$$x_j = 0 \text{ or } 1 \text{ for } j = 1, \dots, 7$$

LP Formulation Summary

Max

$0.486x_1 + 1.08x_2 + 2.2x_3 + 1.428x_4 + 1.575x_5 + 0.282x_6 + 4.224x_7$

s.t.

$6x_1 + 12x_2 + 20x_3 + 14x_4 + 15x_5 + 2x_6 + 32x_7 \leq 45$
 $125x_1 + 150x_2 + 200x_3 + 40x_4 + 40x_5 + 20x_6 + 100x_7 \leq 420$
 $x_1 + x_2 - x_3 \geq 0$
 $x_4 + x_5 \leq 1$

$x_j = 0 \text{ or } 1 \text{ for } j = 1 \dots 7$

	TV/VCRs	TVs	Projection TV	VCRs	DVD Players	Video Players	Home Computers			
Constraints	X1	X2	X3	X4	X5	X6	X7	LHS		RHS
Initial Investment	6	12	20	14	15	2	32	41	<=	45
Floor Space (Sq.Ft)	125	150	200	40	40	20	100	365	<=	420
#3	1	1	-1	0	0	0	0	0	>=	0
#4	0	0	0	1	1	0	0	1	<=	1
#5	0	1	0	0	0	-1	0	0	>=	0
#6	1	1	1	1	1	1	1	3	>=	3
Exp Rate of Return	8.10%	9%	11%	10.20%	10.50%	14.10%	13.20%			
Expected Return	0.486	1.08	2.2	1.428	1.575	0.282	4.224			
					Optimal Solution					
Objective value (max)		4.261			Introduce: TV/VCRs, Projection TVs, and DVD Players					
					Do Not Introduce: TVs, VCRs, Video Games, and Home Computers					
					Total Expected Return: \$4,261					
Decision variables	X1	X2	X3	X4						
	1	0	1	0	1	0	0			

Transportation Problem

□ Linear Programming Formulation

x_{ij} = number of units shipped from origin i to destination j

c_{ij} = cost per unit of shipping from origin i to destination j

s_i = supply or capacity in units at origin i

d_j = demand in units at destination j

$$\text{Min } \sum_{i=1}^m \sum_{j=1}^n c_{ij} x_{ij}$$

s. t.

$$\sum_{j=1}^n x_{ij} \leq s_i \quad i = 1, \dots, m \quad \text{Supply}$$

$$\sum_{i=1}^m x_{ij} = d_j \quad j = 1, \dots, n \quad \text{Demand}$$

$$x_{ij} \geq 0 \quad \text{for all } i \text{ and } j$$

Transportation Problem

LP Formulation Special Cases

– The objective is maximizing profit or revenue:
Solve as a maximization problem.

– Minimum shipping guarantee from i to j :

$$x_{ij} \geq L_{ij}$$

– Maximum route capacity from i to j :

$$x_{ij} \leq L_{ij}$$

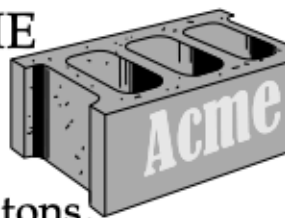
– Unacceptable route:

Remove the corresponding decision variable.

a

Transportation Problem: ACME

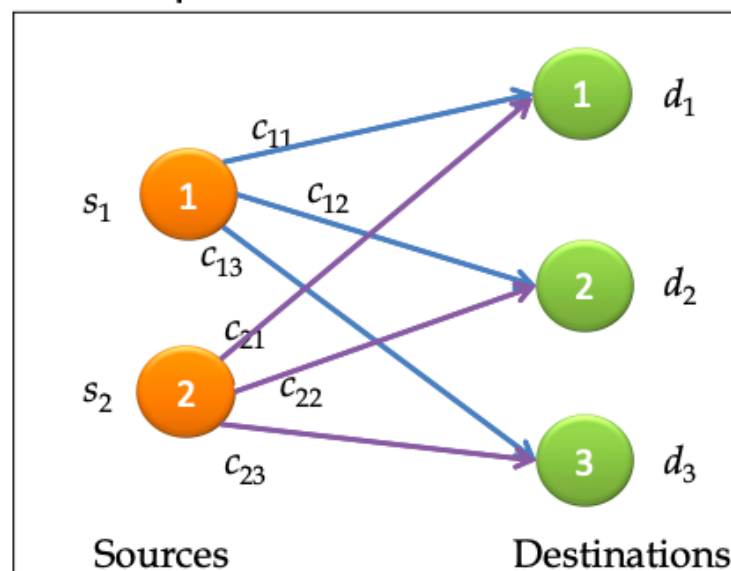
Acme Block Company has orders for 80 tons of concrete blocks at three suburban locations as follows: Northwood -- 25 tons, Westwood -- 45 tons, and Eastwood -- 10 tons.



Acme has two plants, each of which can produce 50 tons per week.

Delivery cost per ton from each plant to each suburban location is shown below. How should end of week shipments be made to fill the above orders?

Network Representation



□ Delivery Cost \$ Per Ton

	<u>Northwood</u>	<u>Westwood</u>	<u>Eastwood</u>
Plant 1	24	30	40
Plant 2	30	40	42

LP Formulation

Decision Variables

	<u>Northwood</u>	<u>Westwood</u>	<u>Eastwood</u>
Plant 1	X_{11}	X_{12}	X_{13}
Plant 2	X_{21}	X_{22}	X_{23}

$$\text{Min } 24 X_{11} + 30 X_{12} + 40 X_{13} + 30 X_{21} + 40 X_{22} + 42 X_{23}$$

s.t.

$$X_{11} + X_{12} + X_{13} \leq 50 \quad \text{Supply at Plant 1}$$

$$X_{21} + X_{22} + X_{23} \leq 50 \quad \text{Supply at Plant 2}$$

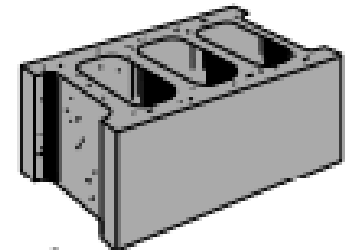
$$X_{11} + X_{21} = 25 \quad \text{Demand at Northwood}$$

$$X_{12} + X_{22} = 45 \quad \text{Demand at Westwood}$$

$$X_{13} + X_{23} = 10 \quad \text{Demand at Eastwood}$$

$$X_{11}, X_{12}, X_{13}, X_{21}, X_{22}, X_{23} \geq 0 \quad \text{Non-negativity}$$

Transportation Problem: ACME



□ Solution

A	B	C	D	E	F	G
Lecture 5: ACME (Transportation Problem)						
	Destination					
Origin	Northwood	Westwood	Eastwood	Supply		
Plant 1	24	30	40	50		
Plant 2	30	40	42	50		
Demand	25	45	10			
Model						
	Min Cost	2490				
	Destination					
Origin	Northwood	Westwood	Eastwood	Total		
Plant 1	5	45	0	50	<=	50
Plant 2	20	0	10	30	<=	50
Total	25	45	10			
	=	=	=			
	25	45	10			

□ Optimal Solution

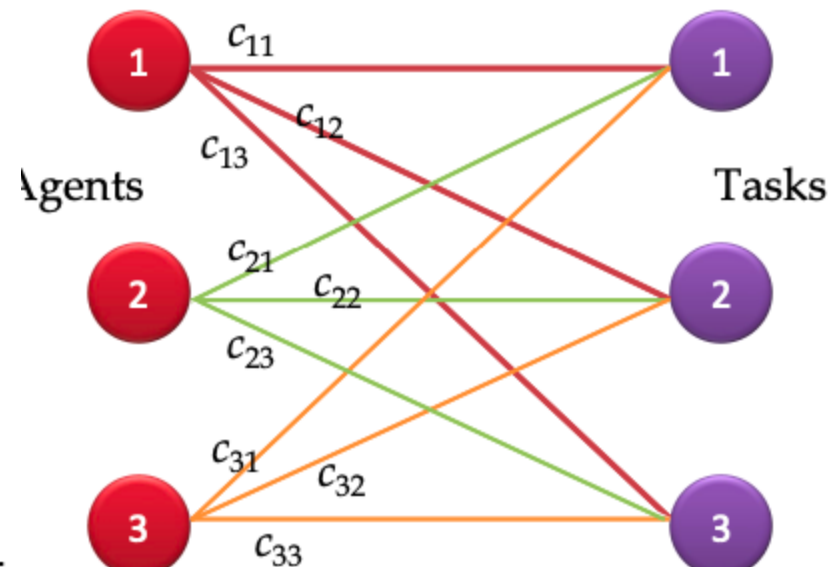
<u>From</u>	<u>To</u>	<u>Amount</u>	<u>Cost</u>
Plant 1	Northwood	5	120
Plant 1	Westwood	45	1,350
Plant 2	Northwood	20	600
Plant 2	Eastwood	10	<u>420</u>

Total Cost = \$2,490

Assignment Problem

- An assignment problem seeks to minimize the total cost assignment of m workers to m jobs, given that the cost of worker i performing job j is c_{ij} .
- It assumes all workers are assigned and each job is performed.
- An assignment problem is a special case of a transportation problem in which all supplies and all demands are equal to 1; hence assignment problems may be solved as linear programs.
- The network representation of an assignment problem with three workers and three jobs is shown on the next slide.

Network Representation



Linear Programming Formulation

- c_{ij} = cost of assigning agent i to task j
- x_{ij} = 1 if agent i is assigned to task j ; = 0 otherwise

$$\begin{aligned}
 &\text{Min} \sum_{i=1}^m \sum_{j=1}^n c_{ij} x_{ij} \\
 &\sum_{j=1}^n x_{ij} \leq 1 \quad i=1,2,\dots,m \quad \text{Agents} \\
 &\sum_{i=1}^m x_{ij} = 1 \quad j=1,2,\dots,n \quad \text{Tasks} \\
 &x_{ij} \geq 0 \quad \text{for all } i \text{ and } j
 \end{aligned}$$

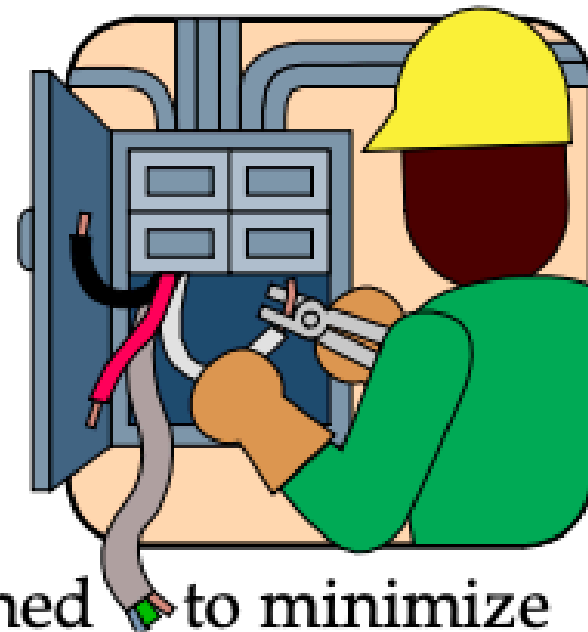
- LP Formulation Special Cases
 - Number of agents exceeds the number of tasks:
 - Extra agents simply remain unassigned.
 - Number of tasks exceeds the number of agents:
 - Add enough dummy agents to equalize the number of agents and the number of tasks. The objective function coefficients for these new variable would be zero.
 - The assignment alternatives are evaluated in terms of revenue or profit:
 - Solve as a maximization problem
 - An assignment is unacceptable:
 - Remove the corresponding decision variable.
 - An agent is permitted to work t tasks:

$$\sum_{j=1}^n x_{ij} \leq t \quad i=1,2,\dots,m \quad \text{Agents}$$

Assignment Problem: Rocky Electronics

Rocky Electrics pays his subcontractors a fixed fee plus mileage for work performed. On a given day the contractor is faced with three electrical jobs associated with various projects. Given below are the distances between the subcontractors and the projects.

<u>Subcontractor</u>	<u>Projects</u>		
	<u>A</u>	<u>B</u>	<u>C</u>
Westside	50	36	16
Federated	28	30	18
Goliath	35	32	20
Universal	25	25	14



How should the contractors be assigned to minimize total mileage costs?

Assignment Problem: Rocky Electronics



□ Linear Programming Formulation

Let $x_{ij} = 1$ if contractor i is assigned to project j , where $i = 1, 2, 3, 4$ and $j = 1, 2, 3$

$$\begin{aligned} \text{Min} \quad & 50x_{11} + 36x_{12} + 16x_{13} + 28x_{21} + 30x_{22} + 18x_{23} \\ & + 35x_{31} + 32x_{32} + 20x_{33} + 25x_{41} + 25x_{42} + 14x_{43} \end{aligned}$$

$$\begin{aligned} \text{s.t.} \quad & x_{11} + x_{12} + x_{13} \leq 1 \\ & x_{21} + x_{22} + x_{23} \leq 1 \\ & x_{31} + x_{32} + x_{33} \leq 1 \\ & x_{41} + x_{42} + x_{43} \leq 1 \\ & x_{11} + x_{21} + x_{31} + x_{41} = 1 \\ & x_{12} + x_{22} + x_{32} + x_{42} = 1 \\ & x_{13} + x_{23} + x_{33} + x_{43} = 1 \\ & x_{ij} = 0 \text{ or } 1 \text{ for all } i \text{ and } j \\ & x_{ij} \geq 0 \text{ for all } i \text{ and } j \end{aligned}$$

- Excel Solution

The optimal assignment is:

	A	B	C	D	E	F	G
1	Rocky Electronics (Assignment Problem)						
2							
3		Tasks					
4	Subcontractor	A	B	C			
5	Westside	50	36	16			
6	Federated	28	30	18			
7	Goliath	35	32	20			
8	Universal	25	25	14			
9							
10							
11	Model						
12							
13	Min mileage costs		69				
14							
15		Tasks					
16	Subcontractor	A	B	C	Total		
17	Westside	0	0	1	1	⇐	1
18	Federated	1	0	0	1	⇐	1
19	Goliath	0	0	0	0	⇐	1
20	Universal	0	1	0	1	⇐	1
21	Total	1	1	1			
22		=	=	=			
23		1	1	1			

<u>Subcontractor</u>	<u>Project</u>	<u>Distance</u>
Westside	C	16
Federated	A	28
Goliath	(unassigned)	
Universal	B	<u>25</u>
Total Distance = 69 miles		

Shortest-Route Problem

The shortest-route problem is concerned with finding the shortest path in a network from one node (or set of nodes) to another node (or set of nodes).

If all arcs in the network have non-negative values then a labeling algorithm can be used to find the shortest paths from a particular node to all other nodes in the network.

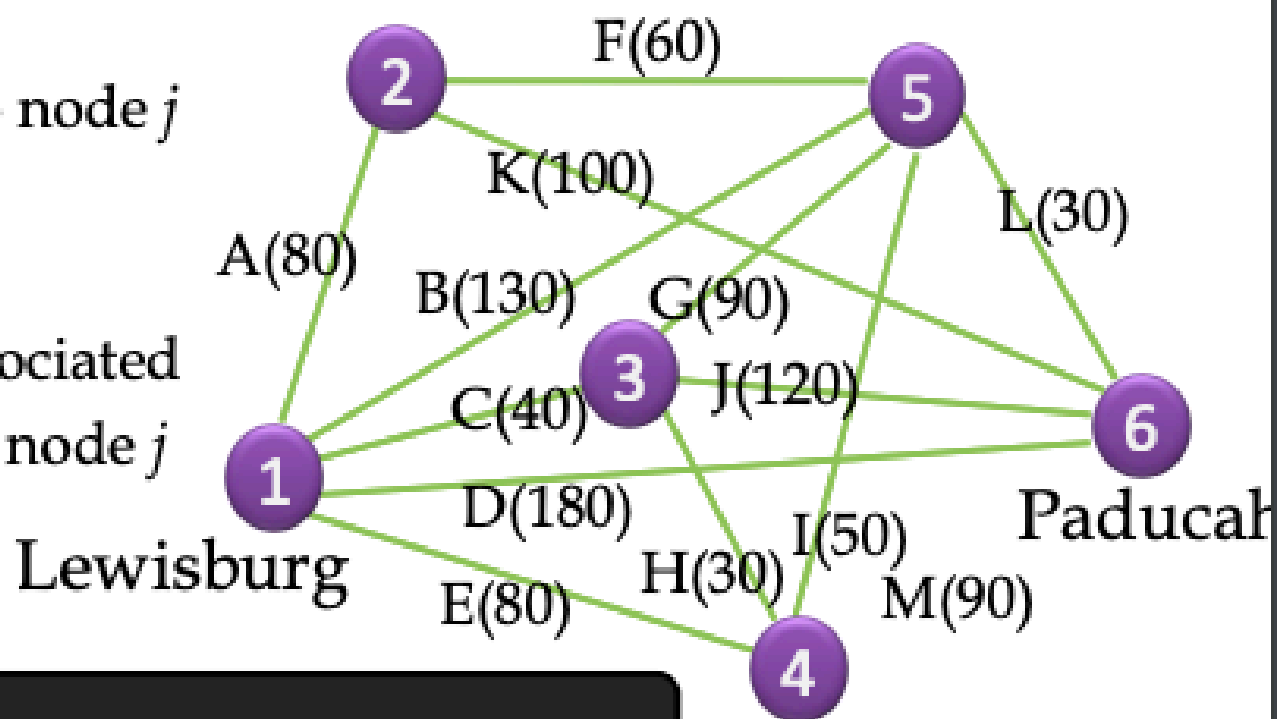
The criterion to be minimized in the shortest-route problem is not limited to distance even though the term "shortest" is used in describing the procedure. Other criteria include time and cost. (Neither time nor cost are necessarily linearly related to distance.)

Linear Programming Formulation

Using the notation:

$$x_{ij} = \begin{cases} 1 & \text{if the arc from node } i \text{ to node } j \\ & \text{is on the shortest route} \\ 0 & \text{otherwise} \end{cases}$$

c_{ij} = distance, time, or cost associated with the arc from node i to node j



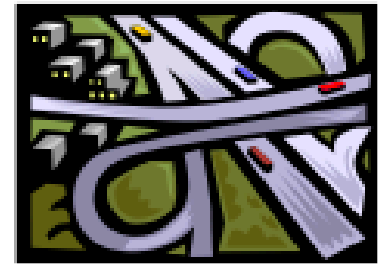
$$\text{Min } \sum_{\text{all arcs}} c_{ij} x_{ij}$$

$$\text{s.t. } \sum_{\text{arcs out}} x_{ij} = 1 \quad \text{Origin node } i$$

$$\sum_{\text{arcs out}} x_{ij} - \sum_{\text{arcs in}} x_{ij} = 0 \quad \text{Transshipment nodes}$$

$$\sum_{\text{arcs in}} x_{ij} = 1 \quad \text{Destination node } j$$

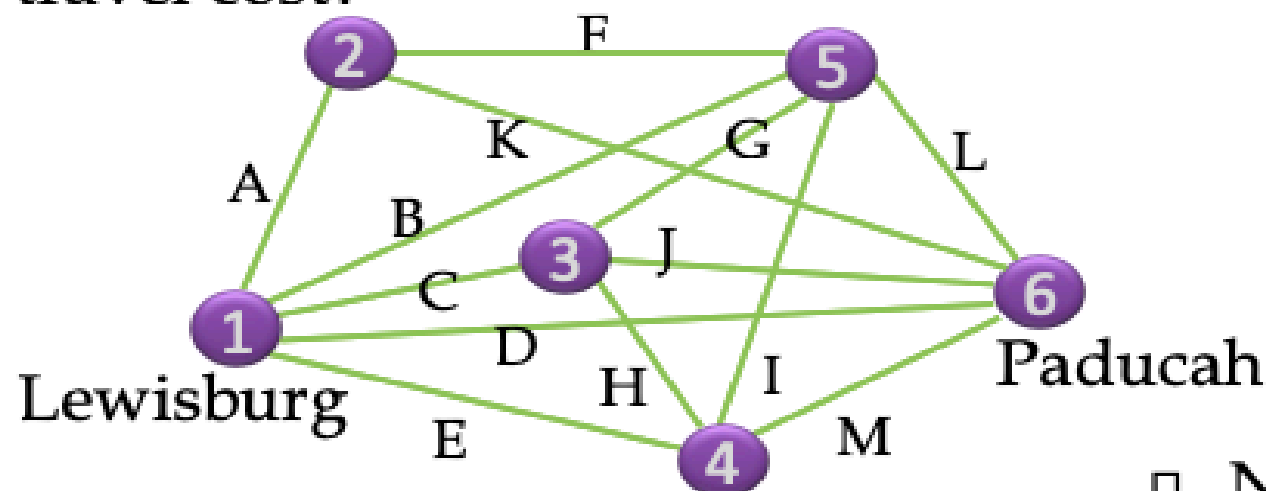
Shortest Route: Winslow



Susan Winslow has an important business meeting in Paducah this evening. She has a number of alternate routes by which she can travel from the company headquarters in Lewisburg to Paducah. The network of alternate routes and their respective travel time, ticket cost, and transport mode appear on the next two slides. If Susan earns a wage of \$15 per hour, what route should she take to minimize the total travel cost?

□ Route Costs

Route	Transport Mode	Time (hours)	Ticket Cost
A	Train	4	\$ 20
B	Plane	1	\$115
C	Bus	2	\$ 10
D	Taxi	6	\$ 90
E	Train	3 1/3	\$ 30
F	Bus	3	\$ 15
G	Bus	4 2/3	\$ 20
H	Taxi	1	\$ 15
I	Train	2 1/3	\$ 15
J	Bus	6 1/3	\$ 25
K	Taxi	3 1/3	\$ 50
L	Train	1 1/3	\$ 10
M	Bus	4 2/3	\$ 20



□ Network Model

Shortest Route: Winslow



□ Network Model with Total Costs

• LP Formulation

$$\begin{aligned} \text{Min } & 80x_{12} + 40x_{13} + 80x_{14} + 130x_{15} + 180x_{16} \\ & + 60x_{25} + 100x_{26} \\ & + 30x_{34} + 90x_{35} + 120x_{36} \\ & + 30x_{43} + 50x_{45} + 90x_{46} \\ & + 60x_{52} + 90x_{53} + 50x_{54} + 30x_{56} \end{aligned}$$

Objective
Function

Route	Transport Mode	Total Cost
A	Train	\$ 80
B	Plane	\$130
C	Bus	\$ 40
D	Taxi	\$180
E	Train	\$ 80
F	Bus	\$ 60
G	Bus	\$ 90
H	Taxi	\$ 30
I	Train	\$ 50
J	Bus	\$120
K	Taxi	\$100
L	Train	\$ 30
M	Bus	\$ 90

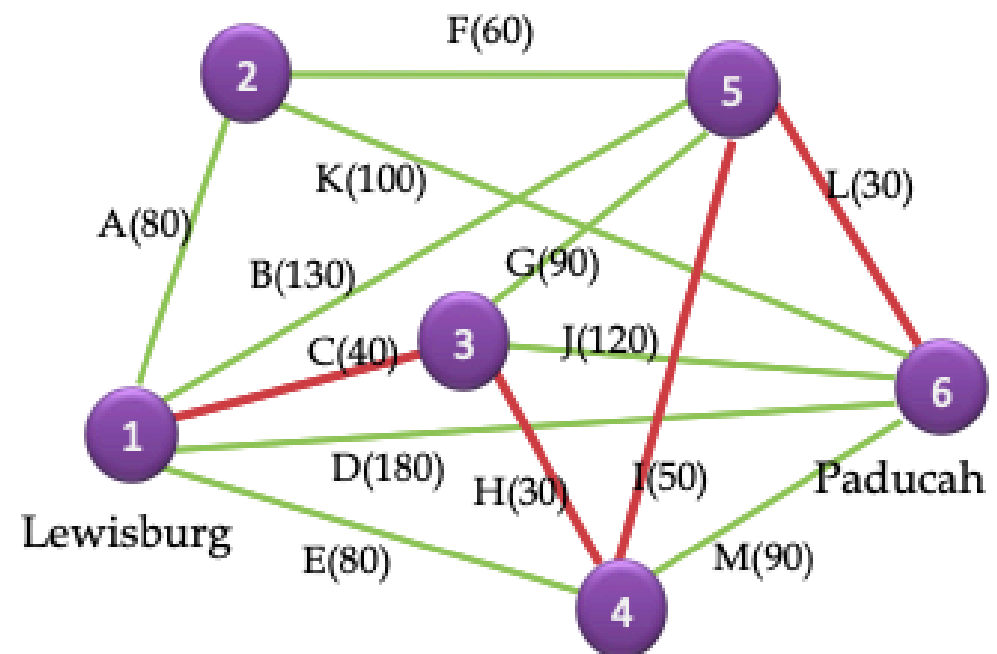
ah

$$\begin{aligned} x_{12} + x_{13} + x_{14} + x_{15} + x_{16} &= 1 && \text{(origin)} \\ -x_{12} + x_{25} + x_{26} - x_{52} &= 0 && \text{(node 2)} \\ -x_{13} + x_{34} + x_{35} + x_{36} - x_{43} - x_{53} &= 0 && \text{(node 3)} \\ -x_{14} - x_{34} + x_{43} + x_{45} + x_{46} - x_{54} &= 0 && \text{(node 4)} \\ -x_{15} - x_{25} - x_{35} - x_{45} + x_{52} + x_{53} + x_{54} + x_{56} &= 0 && \text{(node 5)} \\ x_{16} + x_{26} + x_{36} + x_{46} + x_{56} &= 1 && \text{(destination)} \end{aligned}$$

Node Flow
Conservation
Constraints

• Solution Shortest Route: Winslow

	Arc		Cost	Decision Variable	On Shortest Route?
	Start Node	End Node			
A	1	2	80	0	$x_{13} = 1$ Yes
B	1	5	130	0	
C	1	3	40	1	
D	1	6	180	0	
E	1	4	80	0	
F	2	5	60	0	$x_{34} = 1$ Yes
G	3	5	90	0	
H	3	4	30	1	
I	4	5	50	1	$x_{45} = 1$ Yes
J	3	6	120	0	
K	2	6	100	0	$x_{56} = 1$ Yes
L	5	6	30	1	
M	4	6	90	0	
				0	
A-Rev	2	1	80	0	
B-Rev	5	1	130	0	
C-Rev	3	1	40	0	
D-Rev	6	1	180	0	
E-Rev	4	1	80	0	
F-Rev	5	2	60	0	
G-Rev	5	3	90	0	
H-Rev	4	3	30	0	
I-Rev	5	4	50	0	
J-Rev	6	3	120	0	
K-Rev	6	2	100	0	
L-Rev	6	5	30	0	
M-Rev	6	4	90	0	



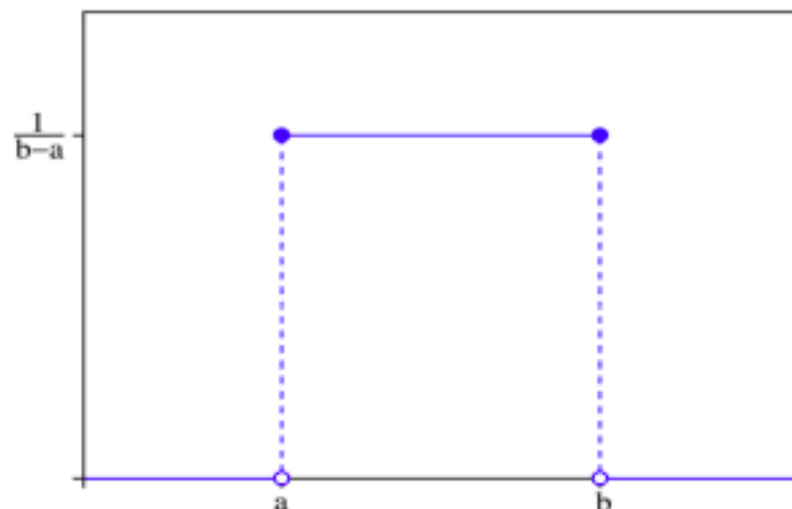
Node	Arcs Used		Net		Balance
	Out	In			
1	1	0	1	=	1
2	0	0	0	=	0
3	1	1	0	=	0
4	1	1	0	=	0
5	1	1	0	=	0
6	0	1	-1	=	-1
Total Costs			150		

Minimum total cost = \$150

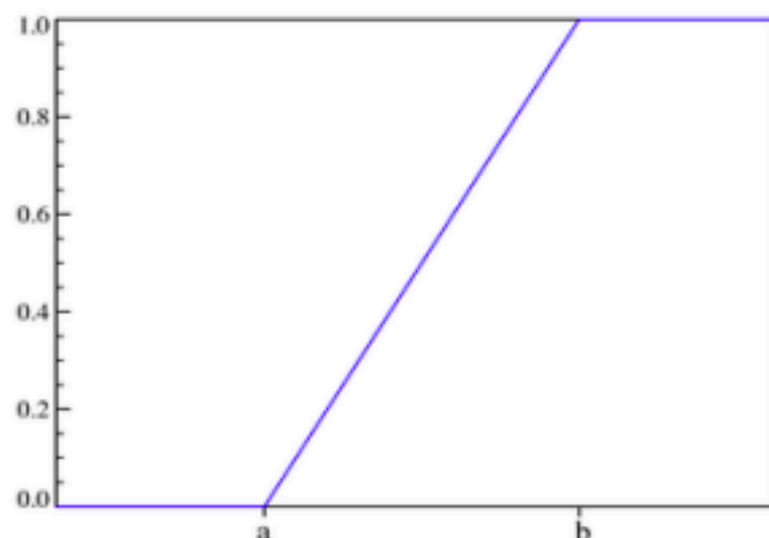
Costs							Arcs									
-->	1	2	3	4	5	6	-->	1	2	3	4	5	6			
1	999	80	40	80	130	180	1	0	0	1	0	0	0	1	1 =	1
2	80	999	999	999	60	100	2	0	0	0	0	0	0	0	0 =	0
3	40	999	999	30	90	120	3	0	0	0	1	0	0	1	0 =	0
4	80	999	30	999	50	90	4	0	0	0	0	1	0	1	0 =	0
5	130	60	90	50	999	30	5	0	0	0	0	0	1	1	0 =	0
6	180	100	120	90	30	999	6	0	0	0	0	0	0	0	-1 =	-1

Continuous Probability Distributions: Uniform Distribution

- Similar to discrete case
- Except that now if $X \sim U[a, b]$ then X can take all values within a and b including them
- A probability distribution function (pdf) normally written as $f(x)$
- A cumulative distribution function (cdf) written as $F(x)$; $F(x) = \Pr(X \leq x)$



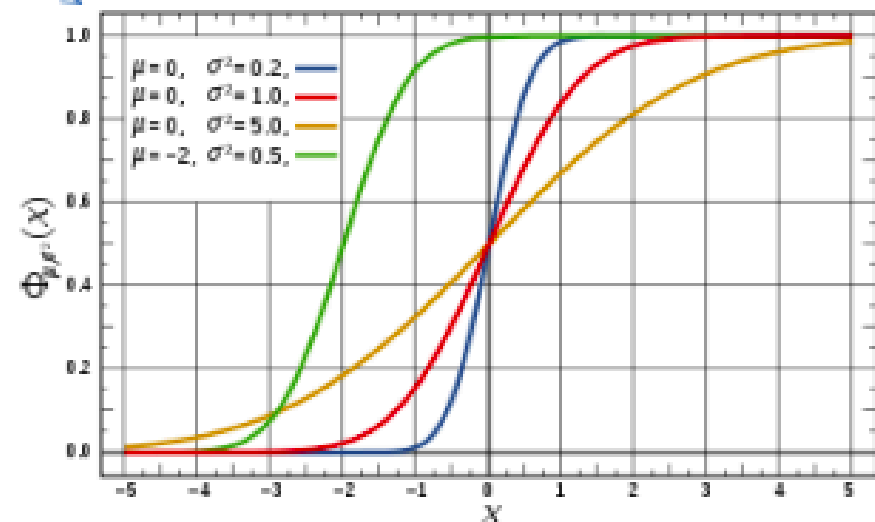
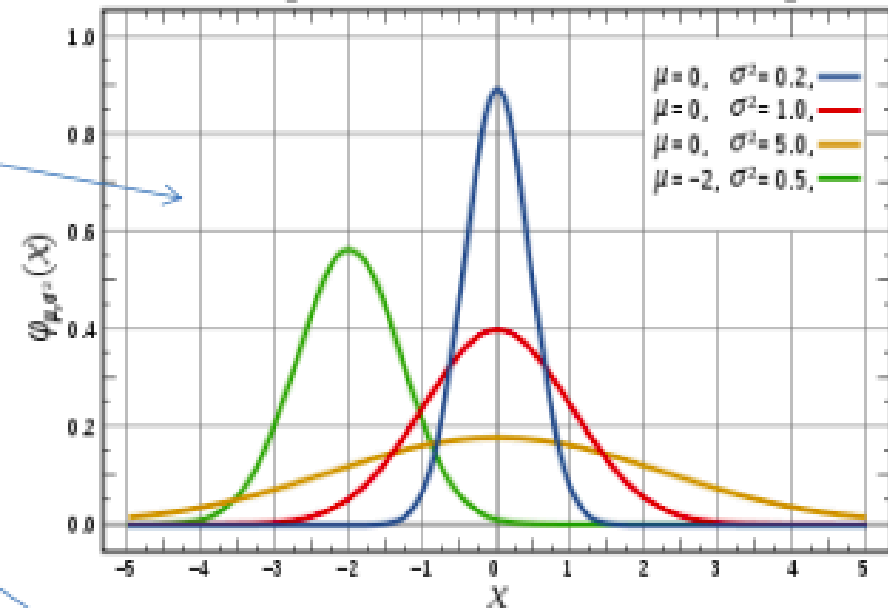
$$f(x) = 1/(b-a) \text{ for all } a \leq x \leq b$$



$$F(x) = (x - a) / (b - a) \text{ for all } a \leq x \leq b$$

Continuous Probability Distributions: Normal Distribution (Continued)

- Pdf can be obtained by the Excel formula `NORMDIST(x,  $\mu$ ,  $\sigma$ , 0)`
- Cdf of a normal distribution can be obtained by the Excel formula `NORMDIST(x,  $\mu$ ,  $\sigma$ , 1)`
- For a standard normal distribution $\mu = 0$, $\sigma = 1$



Continuous Probability Distributions: Exponential Distribution

A random variable with this distribution takes on only positive values

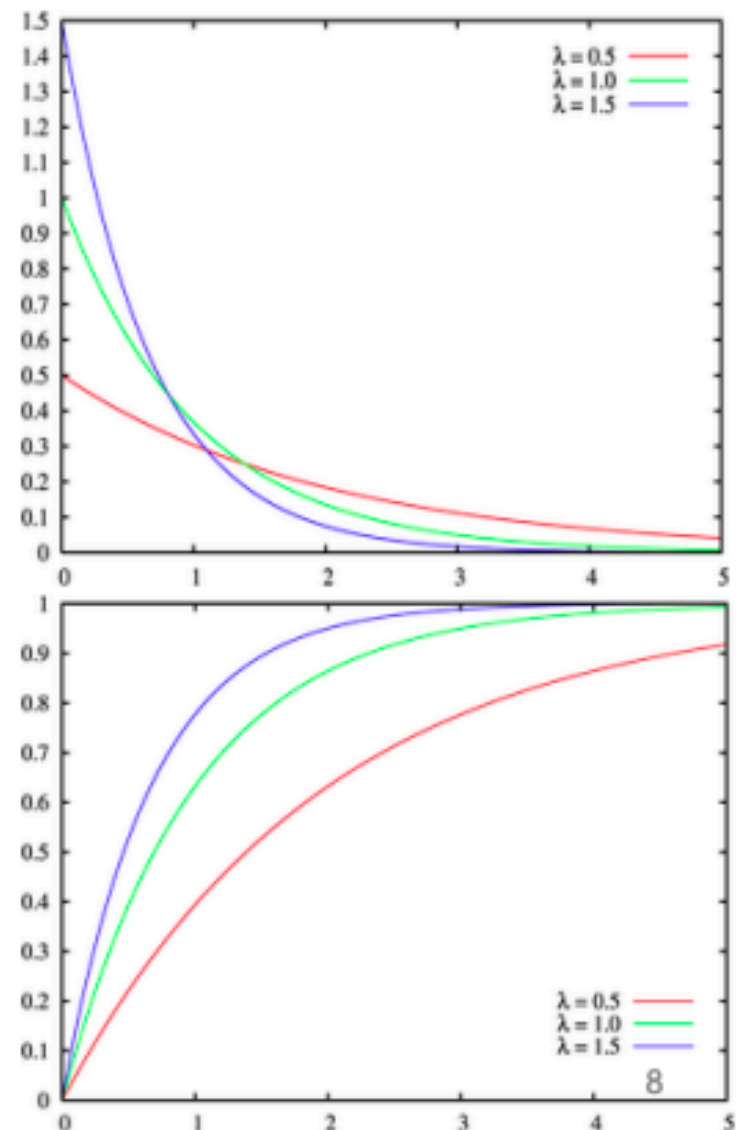
Has only a single parameter: rate λ

The pdf is given by $f(x) = \lambda e^{-\lambda x}$

The cdf is given by $F(x) = 1 - e^{-\lambda x}$

Has applications mostly in queuing systems

- E.g., time it takes to provide treatment to a patient at a hospital, inter-arrival time between two customers at a restaurant



- If $X \sim \text{Exp}(30)$ then

$$F(x) = 1 - e^{-30x}$$

- How to find F^{-1} (read as “F-inverse”)?

- Suppose $F(x) = y$. Then $x = F^{-1}(y)$
- So $1 - e^{-30x} = y \Rightarrow e^{-30x} = 1 - y$
- $\Rightarrow x = -\frac{\ln(1-y)}{30}$

- So to get X that is exponentially distributed with rate 30, use the formula

$$-\frac{\ln(1-\text{RAND}())}{30}$$

Suppose X is normally distributed with mean 0 and standard deviation 1

How to generate the random variable X in Excel?

Need to find inverse function for Normal distribution

- Excel has a function NORMINV() which does that

So X can be generated in Excel by
NORMINV(RAND(),0,1)

Model Simulation Using Random Events

RAND() can be used for generating random numbers, i.e. random variables that are uniformly distributed between 0 and 1

We also saw examples of generating r.v.'s with *some specific distributions*

How to generate random variable X with a *general* distribution that has a cdf $F(x)$?

- E.g., $X \sim \text{Exp}(30)$

Inverse transformation technique (provided it exists)

- Uses the fact that $F(X) \sim U[0, 1]$
- So to get X we just find the inverse function F^{-1} and compute $F^{-1}(U[0, 1])$, i.e., $F^{-1}(\text{RAND}())$ using Excel

Discrete Event Simulation

- Consider the Excel table below.
- Repetitive evaluation of `VLOOKUP(RAND(),$J$7:$K$9,2,1)` will produce a sequence of randomly chosen values:
4,6,5,5,4,4,5,4,6,5,6,5,4,4,5,5,5,5,...

	J	K
4		
5		Number
6	Cumulative	of Days
7	0	4
8	0.25	5
9	0.75	6
10		

- Make sure the cumulative probability column is sorted in increasing manner.
- Random events of almost any kind of cumulative distribution can be generated using this method.

Heavy Duty Company: Corrective Maintenance versus Preventive Maintenance

The Heavy Duty Company has just purchased a large machine for a new production process

The machine is powered by a motor that occasionally breaks down and requires a major overhaul. Therefore, a second standby motor is kept, and the two motors are rotated in use

The breakdowns always occur on the fourth, fifth, or sixth day that the motor is in use. Fortunately, it takes fewer than three days to overhaul a motor, so a replacement is always ready

Cost of a Replacement Cycle that Begins with a Breakdown	
Replace a Motor	\$2,000
Lost production during replacement	5,000
Overhaul a motor	<u>4,000</u>
Total	\$11,000

Probability Distribution of Breakdowns

Day	Probability of a Breakdown	Corresponding Random Numbers
1, 2, 3	0	
4	0.25	0.0000 to 0.2499
5	0.5	0.2500 to 0.7499
6	0.25	0.7500 to 0.9999

Corrective Maintenance

Replace a motor only when it fails

Preventive maintenance

Replace a motor once every three days, i.e., before there is any failure

Question

Of the two, which is better if we consider the cost of repair per day?

Computer Simulation of Corrective Maintenance

Heavy Duty Company Corrective Maintenance Simulation									
	Random	Time Since Last	Cumulative		Cumulative		Distribution of		
Breakdown	Number	Breakdown	Day	Cost	Cost		Time Between Breakdowns		
1	0.2970	5	5	\$11,000	\$11,000				Number
2	0.8017	6	11	\$11,000	\$22,000		Probability	Cumulative	of Days
3	0.5915	5	16	\$11,000	\$33,000		0.25	0	4
4	0.4599	5	21	\$11,000	\$44,000		0.5	0.25	5
5	0.7836	6	27	\$11,000	\$55,000		0.25	0.75	6
6	0.5114	5	32	\$11,000	\$66,000				
7	0.9858	6	38	\$11,000	\$77,000		Breakdown Cost	\$11,000	
8	0.0080	4	42	\$11,000	\$88,000				
9	0.1401	4	46	\$11,000	\$99,000				
10	0.6491	5	51	\$11,000	\$110,000				
26	0.9900	6	133	\$11,000	\$286,000				
27	0.2849	5	138	\$11,000	\$297,000				
28	0.0021	4	142	\$11,000	\$308,000				
29	0.5842	5	147	\$11,000	\$319,000				
30	0.7000	5	152	\$11,000	\$330,000				
							Average Cost per Day		
							\$2,171		

Preventive Maintenance

Preventive maintenance would involve *scheduling* the motor to be removed (and replaced) for an overhaul at a certain time when a breakdown has not occurred

The motor is replaced once every three days during overtime

Since the motor is replaced during overtime, no production is lost

Cost of a Replacement Cycle that Begins without a Breakdown	
Replace a motor on overtime	\$3,000
Lost production during replacement	0
Overhaul a motor before a breakdown	<u>3,000</u>
Total	\$6,000

**Maintenance: Replace Motor
After Three Days (No Simulation needed)**

However company now wants to consider two more "hybrid" options

- Replace motor after four days
- Replace motor after five days

of a replacement cycle is \$6,000 (Replacement always occurs without a breakdown)

Replacement cycle occurs every three days.

$E(\text{Cost per day}) = (\$6,000) / (3 \text{ days}) = \$2,000 \text{ per day.}$

Note that we did not simulate to derive at the result

So preventive maintenance is better than corrective maintenance

Hybrid Policy: Replace Motor After Four Days

Heavy Duty Company Preventive Maintenance Simulation (Replace After 4 Days)										
	Random	Time Until	Scheduled Time	Event That	Cumulative		Cumulative		Distribution of	
Cycle	Number	Breakdown	Until Replacement	Concludes Cycle	Day	Cost	Cost		Time Between Breakdowns	Number of Days
1	0.5547	5	4	Replacement	4	\$6,000	\$6,000			
2	0.2227	4	4	Breakdown	8	\$11,000	\$17,000		Probability	Cumulative
3	0.2842	5	4	Replacement	12	\$6,000	\$23,000		0.25	0
4	0.8486	6	4	Replacement	16	\$6,000	\$29,000		0.5	0.25
5	0.9849	6	4	Replacement	20	\$6,000	\$35,000		0.25	0.75
6	0.6420	5	4	Replacement	24	\$6,000	\$41,000			4
7	0.4254	5	4	Replacement	28	\$6,000	\$47,000		Breakdown Cost	\$11,000
8	0.8827	6	4	Replacement	32	\$6,000	\$53,000		Replacement Cost	\$6,000
9	0.5067	5	4	Replacement	36	\$6,000	\$59,000			
10	0.6083	5	4	Replacement	40	\$6,000	\$65,000		Replace After	4
11	0.1119	4	4	Breakdown	44	\$11,000	\$76,000			days
12	0.7532	6	4	Replacement	48	\$6,000	\$82,000			
13	0.0810	4	4	Breakdown	52	\$11,000	\$93,000			
14	0.9176	6	4	Replacement	56	\$6,000	\$99,000			
15	0.9421	6	4	Replacement	60	\$6,000	\$105,000			
16	0.2996	5	4	Replacement	64	\$6,000	\$111,000			
17	0.2628	5	4	Replacement	68	\$6,000	\$117,000			
18	0.5714	5	4	Replacement	72	\$6,000	\$123,000			
19	0.8221	6	4	Replacement	76	\$6,000	\$129,000			
20	0.3451	5	4	Replacement	80	\$6,000	\$135,000			
21	0.9246	6	4	Replacement	84	\$6,000	\$141,000			
22	0.8336	6	4	Replacement	88	\$6,000	\$147,000			
23	0.2363	4	4	Breakdown	92	\$11,000	\$158,000			
24	0.2234	4	4	Breakdown	96	\$11,000	\$169,000			
25	0.5370	5	4	Replacement	100	\$6,000	\$175,000			
26	0.1053	4	4	Breakdown	104	\$11,000	\$186,000			
27	0.2907	5	4	Replacement	108	\$6,000	\$192,000			
28	0.9929	6	4	Replacement	112	\$6,000	\$198,000			
29	0.4430	5	4	Replacement	116	\$6,000	\$204,000		Average Cost per Day	
30	0.0148	4	4	Breakdown	120	\$11,000	\$215,000			\$1,792

Hybrid Policy: Replace Motor After Five Days

Heavy Duty Company Preventive Maintenance Simulation (Replace After 5 Days)											
	Cycle	Random Number	Time Until Breakdown	Scheduled Time Until Replacement	Event That Concludes Cycle	Cumulative Day	Cost	Cumulative Cost		Distribution of Time Between Breakdowns	
	1	0.2507	5	5	Breakdown	5	\$11,000	\$11,000			
	2	0.0533	4	5	Breakdown	9	\$11,000	\$22,000		Probability	Number of Days
	3	0.2220	4	5	Breakdown	13	\$11,000	\$33,000		0.25	4
	4	0.9537	6	5	Replacement	18	\$6,000	\$39,000		0.5	5
	5	0.4344	5	5	Breakdown	23	\$11,000	\$50,000		0.25	6
	6	0.9308	6	5	Replacement	28	\$6,000	\$56,000			
	7	0.3795	5	5	Breakdown	33	\$11,000	\$67,000		Breakdown Cost	\$11,000
	8	0.1647	4	5	Breakdown	37	\$11,000	\$78,000		Replacement Cost	\$6,000
	9	0.0821	4	5	Breakdown	41	\$11,000	\$89,000			
	10	0.2078	4	5	Breakdown	45	\$11,000	\$100,000		Replace After	5 days
	11	0.2523	5	5	Breakdown	50	\$11,000	\$111,000			
	12	0.1101	4	5	Breakdown	54	\$11,000	\$122,000			
	13	0.0508	4	5	Breakdown	58	\$11,000	\$133,000			
	14	0.8520	6	5	Replacement	63	\$6,000	\$139,000			
	15	0.1954	4	5	Breakdown	67	\$11,000	\$150,000			
	16	0.1630	4	5	Breakdown	71	\$11,000	\$161,000			
	17	0.8398	6	5	Replacement	76	\$6,000	\$167,000			
	18	0.3896	5	5	Breakdown	81	\$11,000	\$178,000			
	19	0.4729	5	5	Breakdown	86	\$11,000	\$189,000			
	20	0.4233	5	5	Breakdown	91	\$11,000	\$200,000			
	21	0.3360	5	5	Breakdown	96	\$11,000	\$211,000			
	22	0.4046	5	5	Breakdown	101	\$11,000	\$222,000			
	23	0.2413	4	5	Breakdown	105	\$11,000	\$233,000			
	24	0.4437	5	5	Breakdown	110	\$11,000	\$244,000			
	25	0.6097	5	5	Breakdown	115	\$11,000	\$255,000			
	26	0.7479	5	5	Breakdown	120	\$11,000	\$266,000			
	27	0.1329	4	5	Breakdown	124	\$11,000	\$277,000			
	28	0.7997	6	5	Replacement	129	\$6,000	\$283,000			
	29	0.2441	4	5	Breakdown	133	\$11,000	\$294,000		Average Cost per Day	
	30	0.3131	5	5	Breakdown	138	\$11,000	\$305,000		\$2,210	

Time Horizons for Forecasting

Long Range

Medium Range

Short Range

- Mostly only qualitative
- Accurate predictions difficult to obtain
- Used for predicting the future of the economy, long-run future of the market, etc.
- Useful for developing long-range strategic plans

- More accurate than long range forecasting
- Useful for making investment decisions, asset acquisitions, etc.

- Most accurate
- For operational planning and decisions in production, personnel, inventory, logistics, etc.
- Accuracy of forecasts is very critical
- Our discussion mostly focuses on short range forecasting

Time-Series Analysis

Last Value Method

Period	1	2	3	4	5	6	7	8
Actual Demand	1000	1200	1900	800	1120	1350	?	?

- Here, we already know
 $X_1 = 1000, X_2 = 1200, \dots, X_6 = 1350$
- Then, we have to determine F_7 .
- Simply use the last month's actual value.
- That is, $F_{t+1} = X_t$.
- In this example, $F_7 = X_6 = 1350$.
- Reasonable method when conditions tend to change so quickly that data before last period are not reliable indicators of future values.

Averaging Method

Period	1	2	3	4	5	6	7	8
Actual Demand	1000	1200	1900	800	1120	1350	?	?

- Use average of all previous values to date.
- That is, $F_{t+1} = (X_1 + \dots + X_t)/(t)$.
- $F_7 = (1000 + \dots + 1350)/6 = 1228.33$.
- Reasonable method when conditions tend to remain so stable that even the earliest value is a reliable indicator of future values.

Exponential Smoothing (ES) Method

Period	1	2	3	4	5	6	7	8
Actual Demand	1000	1200	1900	800	1120	1350	?	?

- Sophisticated version of moving-average, where more recent periods are given more weight.
- Number of periods need not be specified.
- $F_{t+1} = \alpha X_t + (1-\alpha)F_t$, where $0 \leq \alpha \leq 1$
- Here, α is called the smoothing constant.

Moving Average Method

Period	1	2	3	4	5	6	7	8
Actual Demand	1000	1200	1900	800	1120	1350	?	?

- Middle ground between the previous two methods.
- The number of periods must be specified.
- If 3 periods, $F_7 = (800+1120+1350)/3 = 1090$.
- Reasonable method when conditions tend to change occasionally but not extremely rapidly.

- Example: Let $\alpha = 0.2$, and $F_1 = 1000$.

$$F_2 = 0.2X_1 + 0.8F_1 \\ = 0.2(1000) + 0.8(1000) = 1000$$

$$F_3 = 0.2(1200) + 0.8(1000) = 1040$$

$$F_4 = 0.2(1900) + 0.8(1040) = 1212$$

$$F_5 = 0.2(800) + 0.8(1212) = 1129.6$$

$$F_6 = 0.2(1120) + 0.8(1129.6) = 1127.68$$

$$F_7 = 0.2(1350) + 0.8(1127.68) = 1172.14$$

Note:

- Exponential smoothing is like a weighted moving average with exponentially decreasing weights for all previous periods.

Time-Series Analysis

ES With Trend Method

- In ES, the baseline demand was estimated using a smoothing constant α .
- In ES with trend, we also estimate the trend (whether upward or downward) using a smoothing constant β , $0 \leq \beta \leq 1$.
- Basic idea:

ES With Trend Method

- Smoothing of Baseline Estimate B_{t+1} :
Forecast = Baseline Estimate + Trend Estimate
(where Trend Estimate can be positive or negative)

$$B_{t+1} = \alpha X_t + (1-\alpha)F_t$$
$$= \alpha(\text{latest value}) + (1-\alpha)(\text{latest forecast})$$

- Smoothing of Trend Estimate T_{t+1} :

$$T_{t+1} = \beta(B_{t+1} - B_t) + (1-\beta)T_t$$
$$= \beta(\text{latest trend}) + (1-\beta)(\text{latest trend estimate})$$

- Generating the Forecast:

$$F_{t+1} = B_{t+1} + T_{t+1}.$$

ES With Trend Method

Period	1	2	3	4	5	6	7	8
Actual Demand	1000	1200	1900	800	1120	1350	?	?

- Let $\alpha = 0.2$, $\beta = 0.1$, $B_1 = 1000$, $T_1 = 0$, $F_1 = 1000$

- $B_2 = 0.2(1000) + 0.8(1000) = 1000$,

$$T_2 = 0.1(1000 - 1000) + 0.9(0) = 0$$

$$F_2 = 1000 + 0 = 1000$$

- $B_3 = 0.2(1200) + 0.8(1000) = 1040$,

$$T_3 = 0.1(1040 - 1000) + 0.9(0) = 4$$

$$F_3 = 1040 + 4 = 1044$$

$$B_4 = 0.2(1900) + 0.8(1044) = 1215.2,$$

$$T_4 = 0.1(1215.2 - 1040) + 0.9(4) = 21.12$$

$$F_4 = 1215.2 + 21.12 = 1236.32$$

$$B_5 = 0.2(800) + 0.8(1236.32) = 1149.06,$$

$$T_5 = 0.1(1149.06 - 1215.2) + 0.9(21.12) = 12.39$$

$$F_5 = 1149.06 + 12.39 = 1161.45$$

$$* B_6 = 0.2(1120) + 0.8(1161.72) = 1153.16,$$

$$T_6 = 0.1(1153.16 - 1149.06) + 0.9(12.39) = 11.56$$

$$F_6 = 1153.16 + 11.56 = 1164.72$$

$$* B_7 = 0.2(1350) + 0.8(1164.72) = 1201.78,$$

$$T_7 = 0.1(1201.78 - 1153.16) + 0.9(11.56) = 15.27$$

$$F_7 = 1201.78 + 15.27 = 1217.05$$

Time-Series Analysis

Forecast Accuracy

- Cumulative error
 - Deviations/errors (actual value – forecast) are added up

$$\text{Cumulative Error} = \sum_i (X_i - F_i)$$

- Bias
 - It is the average cumulative error

$$\text{Bias} = \frac{\sum_i (X_i - F_i)}{N}$$

- A large positive value of bias (or cumulative error) indicates the forecast is biased low; a large negative value indicates forecast is biased high.

Better Estimates of Forecast Accuracy

- Mean Absolute Deviation (MAD)
 - Deviations are linearly (proportionately) penalized.

$$MAD = \frac{\sum_i |X_i - F_i|}{N}$$

- Mean Square Error (MSE)
 - Additional deviations are penalized more.

$$MSE = \frac{\sum_i (X_i - F_i)^2}{N}$$

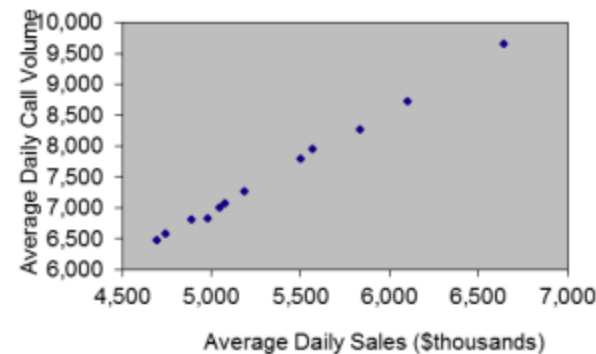
- Root Mean Square Error (RMSE)

$$RMSE = \sqrt{\frac{\sum_i (X_i - F_i)^2}{N}}$$

Causal Forecasting

Sales and Call Volume for a Call Center

	A	B	C	D	E
1	CCW's Average Daily Sales and Call Volume				
2					
3				Sales	Call
4		Year	Quarter	(\$thousands)	Volume
5		1	1	4,894	6,809
6		1	2	4,703	6,465
7		1	3	4,748	6,569
8		1	4	5,844	8,266
9		2	1	5,192	7,257
10		2	2	5,086	7,064
11		2	3	5,511	7,784
12		2	4	6,107	8,724
13		3	1	5,052	6,992
14		3	2	4,985	6,822
15		3	3	5,576	7,949
16		3	4	6,647	9,650



A line trend is clearly visible when call volume is plotted against sales

Causal Forecasting

Causal forecasting obtains a forecast of the quantity of interest (the **response variable**) by relating it directly to one or more other quantities (the **explanatory variables**) that drive the quantity of interest.

Type of Forecasting	Possible Dependent Variable	Possible Independent Variable
Sales	Sales of a product	Amount of advertising
Spare parts	Demand for spare parts	Usage of equipment
Economic trends	Gross domestic product	Various economic factors
Call center call volume	Call volume	Total sales
Any quantity	This same quantity	Time

Causal Forecasting – Linear Regression

- **Linear regression** approximates the relationship between the dependent and the independent variables by a straight line.
- This **linear regression line** is drawn on a graph with the independent variable on the horizontal axis and the dependent variable on the vertical axis.
- The linear regression line has the form: $Y = a + bX$
 - Y = Estimated value of the dependent variable
 - a = Intercept of the linear regression line with the y -axis
 - b = Slope of the linear regression line
 - X = Value of the independent variable
- Our goal is to find estimates for a and b , such that the squared deviations of the predicted values of Y from the actual values X are minimized.
- This is called the **Method of Least Squares (MLS)**.

Causal Forecasting – Linear Regression

- Given the n pairs of past data, a and b are given by:

$$b = \frac{SS_{XY}}{SS_{XX}}, \quad a = \bar{Y} - b\bar{X}$$

where $SS_{XY} = \sum_{i=1}^n \{(X_i - \bar{X})(Y_i - \bar{Y})\}$, $SS_{XX} = \sum_{i=1}^n (X_i - \bar{X})^2$
and $\bar{Y}$ = Mean value of Y, $\bar{X}$ = Mean value of X

CCW's Average Daily Sales and Call Volume								
Year	Quarter	Sales (X) (\$thousands)	Call (Y) Volume	(X-Xm)^2	(Y-Ym)^2	(X-Xm)(Y-Ym)		
1	1	4,894	6,809	219102	518760	337137.0208	n=	12
1	2	4,703	6,465	434391	1132628	701429.4375	SSxy=	6334034
1	3	4,748	6,569	377098	922080	589673.5208	SSxx=	3880177
1	4	5,844	8,266	232244	542801	355052.1042		
2	1	5,192	7,257	28928	74120	46305.1875	b=	1.63240849
2	2	5,086	7,064	76222	216458	128447.7708	a=	-1223.8604
2	3	5,511	7,784	22176	64898	37936.52083		
2	4	6,107	8,724	554901	1427428	889989.1875		
3	1	5,052	6,992	96152	288638	166592.2708		
3	2	4,985	6,822	142192	500203	266692.1875		
3	3	5,576	7,949	45760	176190	89791.52083		
3	4	6,647	9,650	1651011	4497581	2724987.021		
		Xm	Ym					
	Total			3880177	10361782	6334034		
	Mean Values	5,362	7,529					

Causal Forecasting

Excel Template for Linear Regression

BU8401-Lect12-Forecasting.xlsx

=INTERCEPT(D5:D16,C5:C16)

	A	B	C	D	E	F	G	H	I	J
1	Linear Regression of Call Volume vs. Sales Volume for CCW									
2										
3		Time	Independent	Dependent		Estimation	Square		Linear Regression Line	
4		Period	Variable	Variable	Estimate	Error	of Error		y = a + bx	
5		1	4,894	6,809	6,765	43.85	1,923		a =	-1,223.86
6		2	4,703	6,465	6,453	11.64	136		b =	1.63
7		3	4,748	6,569	6,527	42.18	1,780			
8		4	5,844	8,266	8,316	49.93	2,493			
9		5	5,192	7,257	7,252	5.40	29		Estimator	
10		6	5,086	7,064	7,079	14.57	212		If x =	5,000
11		7	5,511	7,784	7,772	11.66	136			
12		8	6,107	8,724	8,745	21.26	452		then y=	6,938.18
13		9	5,052	6,992	7,023	31.07	965			
14		10	4,985	6,822	6,914	91.70	8,408			
15		11	5,576	7,949	7,878	70.55	4,977			
16		12	6,647	9,650	9,627	23.24	540			

We can then use $Y = a + bX$ to estimate Y given any X.

=SLOPE(D5:D16,C5:C16)

Kendall-Lee Notation for Queuing Systems

$A/B/k$ is used to describe various queuing systems where
 A identifies the arrival distribution,
 B the service (departure) distribution and
 k the number of channels for the system.

Symbols used for the arrival and service processes are:

M - Markov distributions (Poisson/exponential),

D - Deterministic (constant) and

G - General distribution (with a known mean and variance).

The Service System

Configuration: How the service system is set up

1. Number of phases (single vs. multiple)



2. Number of servers (channels) (single vs. multiple)



3. Number of waiting lines (single vs. multiple)



Performance of Queuing Systems

L_q : Average queue length

L : Average number of customers in the system

λ : Arrival rate, average number of arrivals per unit time

μ : Service rate, average number of customers served per unit time

ρ : Service utilisation factor (proportion of time server is busy)

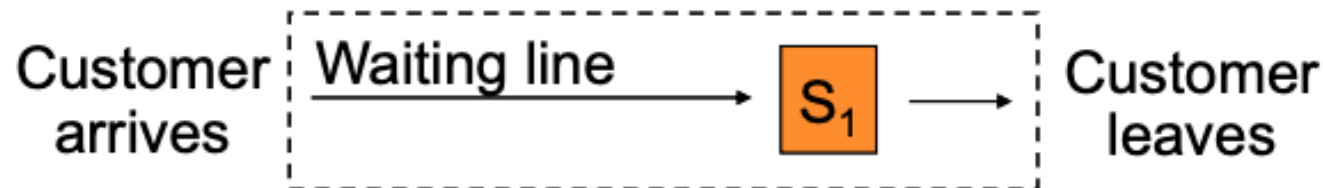
W_q : Average time in queue

W : Average time in system

P_0 : Probability that no one is in the system

P_n : Probability that there are n customers present in the system

Single Server – M/M/1 Queue



- Infinite calling population
- Infinite queue length
- Poisson Arrivals Rates, Exponential inter-arrival times
- FCFS queue discipline, infinite queue size
- Exponential service time
- Customer are patient (i.e., no reneging)

Steady State Measurements for M/M/1

Formulae

Under assumption of $\rho = \frac{\lambda}{\mu} < 1$

$$P_0 = 1 - \frac{\lambda}{\mu} = 1 - \rho$$

$$P_n = P_0 \left(\frac{\lambda}{\mu} \right)^n = P_0 (\rho^n)$$

$$L_q = \frac{\lambda^2}{\mu(\mu - \lambda)} = \frac{\rho^2}{1 - \rho}$$

$$L = L_q + \frac{\lambda}{\mu} = \frac{\frac{\lambda}{\mu}}{1 - \frac{\lambda}{\mu}}$$

$$W_q = \frac{L_q}{\lambda} = \frac{\lambda}{\mu(\mu - \lambda)}$$

$$W = \frac{L}{\lambda} = \frac{1}{\mu - \lambda}$$

[Little's Law]

$$\lambda = 2 \text{ cars per minute} \quad \mu = 4 \text{ cars per minute}$$

$$\rho = \lambda/\mu = 2/4 = 0.5 \quad P(0) = 1 - \rho = 1 - 0.5 = 0.5$$

$$L = \lambda/(\mu - \lambda) = 2/(4 - 2) = 1.0$$

$$L_q = \lambda^2/[\mu(\mu - \lambda)] = 2^2/[4(4 - 2)] = 0.5$$

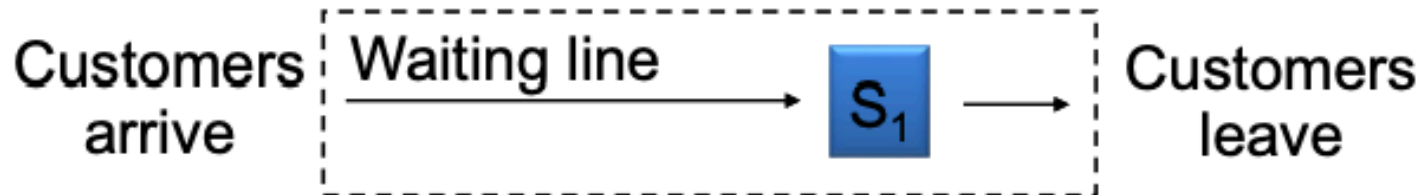
An Example: A

$$W = L/\lambda = 1/2 = 0.5 \text{ min}$$

$$W_q = L_q/\lambda = 0.5/2 = 0.25 \text{ min}$$

A toll road has one attendant at an exit gate. Cars arrive in a Poisson fashion at the rate of 120 cars per hour. It takes the attendant, on the average, 15 seconds to service a car. Service times are exponentially distributed. Obtain the basic queue performance measures for the system. Assume that there is an infinite car calling population and an infinite queue length is possible.

M/G/1 Queue



*Service time is generally distributed with mean $1/\mu$ and standard deviation σ and under assumption of $\rho = \lambda/\mu < 1$

$$P_0 = 1 - \rho$$

$$P_n = P_0 \rho^n$$

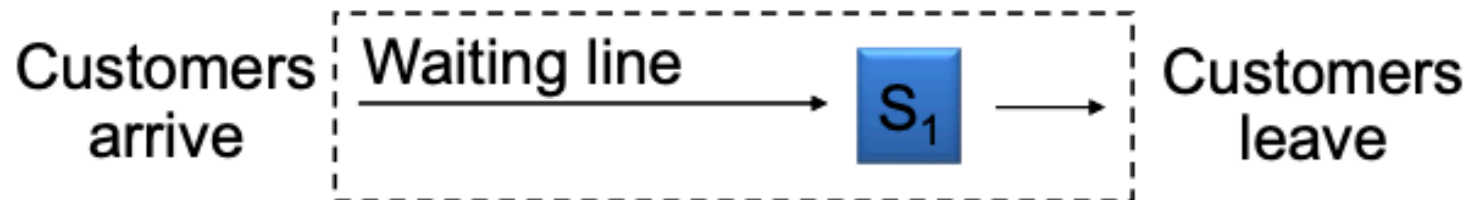
$$L_q = \frac{(\lambda\sigma)^2 + \rho^2}{2(1-\rho)}$$

$$L = L_q + \rho$$

$$W_q = L_q / \lambda$$

$$W = L / \lambda$$

M/D/1 Queue



For Deterministic service time where there is NO variation in service times (ie fixed constant service duration), we set σ to zero in M/G/1 and get:

$$P_0 = 1 - \rho$$

$$P_n = P_0 \rho^n$$

$$L_q = \frac{\rho^2}{2(1-\rho)}$$

$$L = L_q + \rho$$

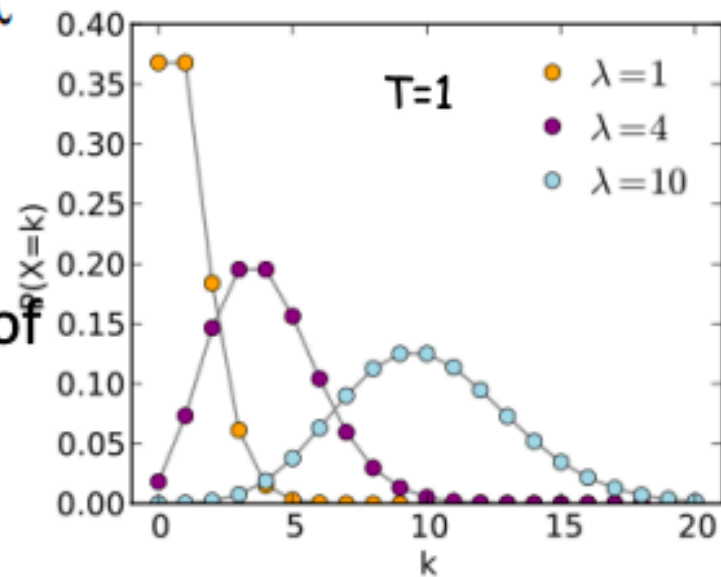
$$W_q = L_q / \lambda$$

$$W = L / \lambda$$

Note: while the formulas look the same, the numerical values differ from those of M/G/1.

The Poisson Distribution

- Arrival rate: λ (on average, there are λ arrivals per unit time)
- If inter-arrival time follows an exponential distribution, the number of arrivals (X) in a period of length T follows a Poisson distribution
- Probability distribution function:



$$P(X = n) = \frac{(\lambda T)^n e^{-\lambda T}}{n!}$$

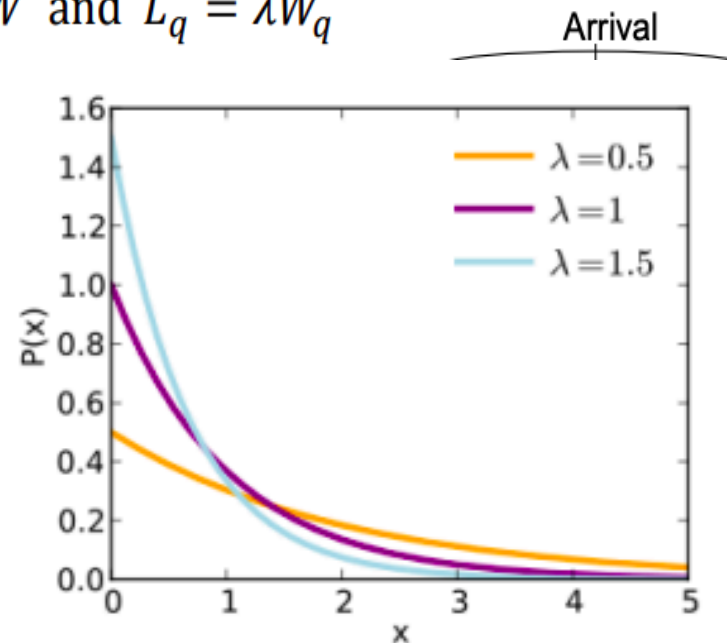
The Exponential

For nearly all queuing systems, there is a relationship between the average time a unit spends in the system or queue and the average number of units in the system or queue.

These relationships, known as Little's flow equations are:

$$L = \lambda W \text{ and } L_q = \lambda W_q$$

- Arrival rate: λ (on average, there are λ arrivals per unit time)
- Inter-arrival time X : time between two consecutive arrivals
- No memory property: future does not depend on past
- Probability density function:



$$f(t) = \lambda e^{-\lambda t}$$

An Example: Joe Ferris

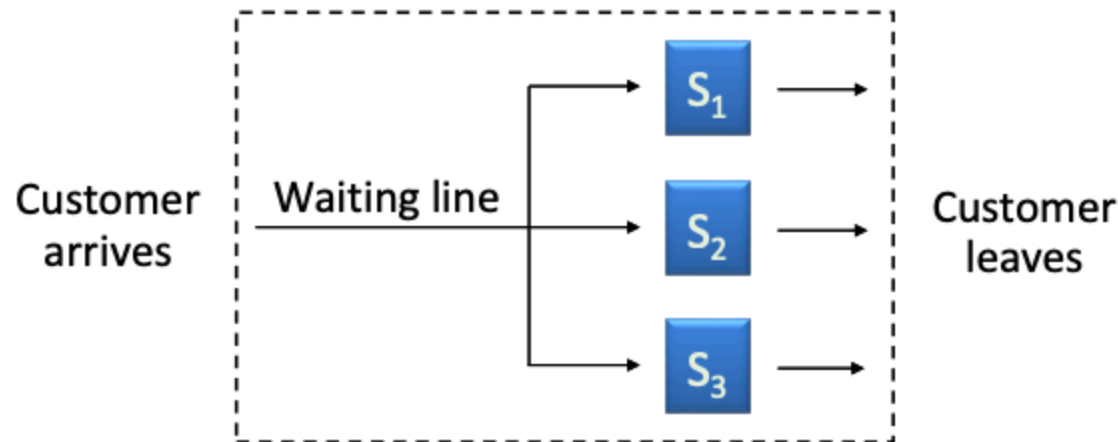
Joe Ferris is a stock trader on the floor of the New York Stock Exchange. Stock transactions arrive at a mean rate of 20 per hour. Each order received by Joe requires an average of two minutes to process.

What is the probability that (a) no orders, (b) exactly 3 orders, and (c) more than 6 orders are received within a 15-minute period?

- Arrival rate for a 15-minute period: $\lambda T = 5$
 - a. $P(x = 0) = (5^0 e^{-5})/0! = e^{-5} = .0067$
 - b. $P(x = 3) = (5^3 e^{-5})/3! = 125(.0067)/6 = .1396$
 - c. $P(x > 6) = 1 - P(X \leq 6)$

Since $P(x = 0) + \dots + P(x = 6) = 0.762$,
 $P(x > 6) = 1 - 0.762 = 0.238$

Multiple Server – M/M/k Queue



- Poisson arrival rate/Exponential service times
 - k multiple identical servers with single queue
 - Each server services at μ customers per unit time
 - Every server is identical in performance
 - Customer does not choose which server for service
-
- $\frac{\lambda}{\mu}$ is called the “implied utilization”, and is often greater than 1.
 - $\rho = \frac{\lambda}{k\mu}$ is our usual “utilization” on a per-server basis. In other words, each server is ρ percent of the time busy.

Multiple Server Model - M/M/k

Formulae

k = number of servers

$k\mu$ = mean effective service rate for the system

which must exceed the arrival rate

$$\lambda < k\mu$$

Probability of no customers in the system (all servers idle) is

$$P_0 = \frac{1}{\left(\sum_{n=0}^{k-1} \frac{1}{n!} \left(\frac{\lambda}{\mu} \right)^n \right) + \frac{1}{k!} \left(\frac{\lambda}{\mu} \right)^k \frac{k\mu}{k\mu - \lambda}}$$

Probability of j customers in the system ($1 \leq j \leq k$):

$$P_j = \left(\frac{\lambda}{\mu} \right)^j \frac{1}{j!} \cdot P_0$$

Probability of j customers in the system ($j \geq k$):

$$P_j = P_k \left(\frac{\lambda}{k\mu} \right)^{j-k} = \left(\frac{\lambda}{k\mu} \right)^j \frac{k^k}{k!} \cdot P_0$$

Multiple Server Model - M/M/k

Average number of customers in queue:

$$L_q = \frac{\lambda \mu \left(\frac{\lambda}{\mu}\right)^k}{(k-1)! (k\mu - \lambda)^2} P_0$$

Average time a customer spends in waiting to be served:

$$W_q = \frac{L_q}{\lambda}$$

Average number of customers in the system:

$$L = L_q + \frac{\lambda}{\mu}$$

Average waiting time in the system (waiting and being served):

$$W = \frac{L}{\lambda}$$

Road Example (Part II)

A toll road has one attendant at an exit gate. Cars arrive in a Poisson fashion at the rate of 120 cars per hour. It takes the attendant, on the average, 15 seconds to service a car. Service times are exponentially distributed. Obtain the basic queue performance measures for the system. Assume that there is an infinite car calling population and an infinite queue length is possible.

Assume now that the arrival rate is 624 per hour and three (instead of one) toll booths are open. What are the performance measures for the new system?

$$\lambda = 10.4/\text{minute}, \mu = 4/\text{minute}, k = 3 \quad \lambda/\mu = 2.60$$

$$P_0 = 0.0345 \text{ (from table)}$$

$$L_q = 4.9328$$

$$W_q = 4.9328/10.4 = 0.4743 \text{ m}$$

$$L = 4.9328 + 2.6 = 7.5328 \quad W = 7.5328/10.4 = 0.7243$$

Road Example (Part III)

What is the LEAST number of attendants that is feasible for this system?

$$\lambda = 10.4/\text{minute}, \mu = 4/\text{minute}, k = ?$$

The LEAST number of attendants is the smallest value of k such that $\lambda < k\mu$.

k	λ	$k\mu$	System OK?
2	10.4	8.0	No
3	10.4	12.0	YES!
4	10.4	16.0	Yes
5	10.4	20.0	Yes

Road Example (Part IV)

If the desirable mean waiting time for cars should be less than two seconds, how many attendants should we have?

$$\lambda = 10.4/\text{minute}, \mu = 4/\text{minute}, k = 3$$

k	P_0	L_q	$W_q = L_q/\lambda$
3	0.034542	4.93281	0.474m = 28.47s
4	0.065148	0.65821	0.063m = 3.80s
5	0.072066	0.16104	0.015m = 0.93s

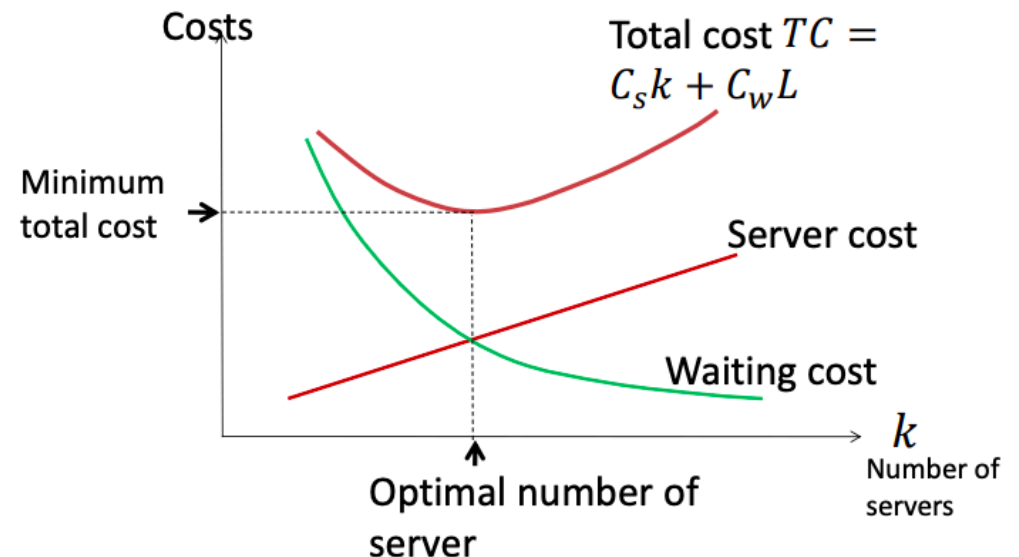
The Total Cost Measure

Cost per hour with c servers (TC)

= Cost of servers + Cost of Waiting
= Number of servers + Expected customers in system
× Cost per server /hour × Cost of waiting per customer /hour

$$TC = C_s \times k + C_w \times L$$

Looking At The Costs



Queuing at American Weavers

American Weavers operates a cloth manufacturing plant which has several weaving machines that jam frequently at the rate of 25 per hour. Machines can be repaired on a FCFS basis by one of the 7 available repair persons. Repair time is exponentially distributed with an average of 15 min. The manager has estimated that the company loses \$100 for every hour that a machine is out of operation. Adding an extra repair person costs \$50 per hour. Determine the optimal number of repair persons.

Queuing Performance Measures

$\lambda = 25/\text{hour}$, $\mu = 4/\text{hour}$, $k = 7$ repair persons

k	No. of Repair Persons				
	7	8	9	10	11
Utilization	0.89	0.78	0.69	0.63	0.57
Exp. Queue Length (Lq)	5.85	1.49	0.54	0.21	0.08
Exp. No. in System (L)	12.10	7.74	6.79	6.46	6.33
Exp. Time in Queue (Wq)	0.23	0.06	0.02	0.01	0.003
Exp. Time in System (W)	0.48	0.31	0.27	0.26	0.25

Balancing the Two Costs

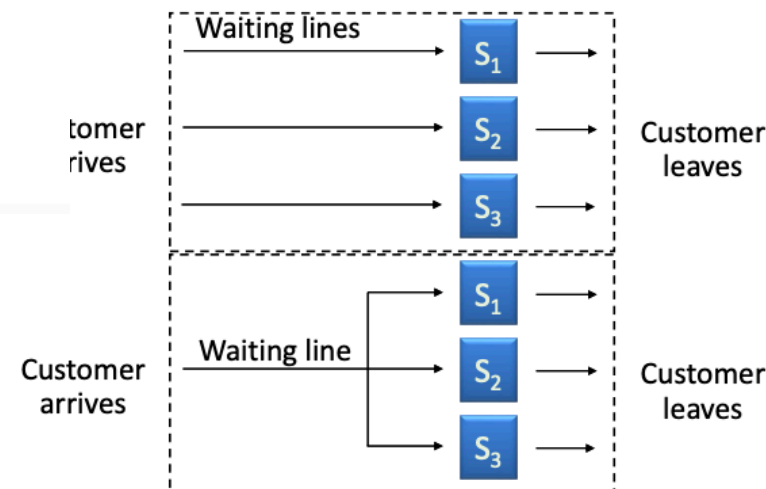
k	No. of Repair Persons				
	7	8	9	10	11
Utilization	0.89	0.78	0.69	0.63	0.57
Exp. Queue Length (L _q)	5.85	1.49	0.54	0.21	0.08
Exp. No. in System (L)	12.10	7.74	6.79	6.46	6.33
Exp. Time in Queue (W _q)	0.23	0.06	0.02	0.01	0.003
Exp. Time in System (W)	0.48	0.31	0.27	0.26	0.25
Hourly Cost (\$)	50x7 + 100 x 12.1	50x8 + 100 x 7.74	50x9 + 100 x 6.79	50x10 + 100 x 6.46	50x11 + 100 x 6.33
=	1560	1174	1129	1146	1183



Waiting Cost

- In most queuing models, the number of customers in ($k \times M/M/1$) system is higher than $M/M/k$ system if both systems have the same L_q (customers per server queue).
- So unless each queue in ($k \times M/M/1$) system is lesser than $M/M/k$ average queue length by at least a factor of $\frac{1}{k}$, $M/M/k$ system will have much lesser customers waiting in queue.

Multiple or Single Queue?



- Same number of servers, but which arrangement is better?

Example – Car Repair Shop

A new tire shop is designed to have 5 workstations in which a car can have 1 to 4 of its tires changed. Regardless of the number of tires to be changed, each workstation takes 15 mins (exponentially distributed) on average to complete its work. Patronizing cars are expected to drive in at a rate of 12 every hour in a Poisson distributed fashion. The shop owner is considering whether to layout the shop in $M/M/5$ or $5 \times M/M/1$.

- If the owner wants to save space allocated for waiting cars, which queuing system should the owner choose?

Example – Car Repair Shop

For M/M/5,

$$\lambda = 12/\text{hr},$$

$$\mu = 4/\text{hr},$$

$$k = 5$$

$$L_q = 0.3542 \text{ cars}$$

$$W_q = 0.0295 \text{ hr} = 1.77 \text{ min}$$

$$L = 3.3542 \text{ cars}$$

$$W = 0.2795 \text{ hr} = 16.78 \text{ min}$$

For each queue of 5xM/M/1,

$$\text{Effective } \lambda = 12/\text{hr} / 5 = 2.4/\text{hr},$$

$$\mu = 4/\text{hr},$$

$$k = 1$$

$$L_q = 0.9 \text{ cars}$$

$$W_q = 0.375 \text{ hr} = 22.5 \text{ min}$$

$$L = 1.5 \text{ cars}$$

$$W = 0.625 \text{ hr} = 37.5 \text{ min}$$

For 5xM/M/1,

$$L_q = 0.9 \times 5 = 4.5 \text{ cars}$$

$$L = 1.5 \times 5 = 7.5 \text{ cars}$$

Example – Car Repair Shop

So for M/M/5,
 $L_q = 0.3542$ cars
 $W_q = 1.77$ min
 $L = 3.3542$ cars
 $W = 16.78$ min

For 5xM/M/1,
 $L_q = 4.5$ cars
 $W_q = 22.5$ min
 $L = 7.5$ cars
 $W = 37.5$ min

So in 5xM/M/1, owner has to set aside about 5 car lots just to queue up waiting customers. Customers are also required to wait for a long time of 22.5 min. If owner configures his workshop to be M/M/5, then only about 1 car lot needs to be set aside. In expensive cities like Singapore, this arrangement saves a lot of money and makes customers wait less.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N
1														
2	lamb =	5.4	p/min											
3	mu =	3	p/min											
4	k =	2												
5	lamb/mu =	1.8												
6				$W_q = \frac{L_q}{\lambda}$		$L_q = \frac{\left(\frac{\lambda}{\mu}\right)^{k+1}}{(k-1)!(k-\lambda/\mu)^2} P_0$			$P_0 = \frac{1}{\left(\sum_{n=0}^{k-1} \frac{1}{n!} \left(\frac{\lambda}{\mu}\right)^n\right) + \frac{1}{k!} \left(\frac{\lambda}{\mu}\right)^k \frac{k\mu}{k\mu-\lambda}}$			$P_j = \left(\frac{\lambda}{\mu}\right)^j \frac{1}{j!} \cdot P_0$		
7					Wq	Lq	k	Term2	P0	P1	P2	P3	P4	Pw
8	n	Term	Sum					j = -->		1	2	3	4	
9	0	1	1				1	-2.25	-0.8					
10	1	1.8	2.8		1.421053	7.673684	2	16.2	0.052632	0.094737	0.085263	0.051158	0.023021	0.852632
11	2	1.62	4.42		0.09854	0.532117	3	2.43	0.145985	0.262774	0.236496	0.141898	0.063854	0.354745
12	3	0.972	5.392		0.019475	0.105164	4	0.795273	0.161622	0.29092	0.261828	0.157097	0.070694	0.128534
13	4	0.4374	5.8294		0.004218	0.02278	5	0.246038	0.164597	0.296275	0.266647	0.159988	0.071995	0.040497
14														
15														
16														
17														
18														
19														

These extra rows of Wq, Lq & k can be used to answer questions like: find least k such that Wq is less than 30 secs (ans: k=3), or Lq is not more than half (ans: k=4).

An Example: Inventory Control at ACT

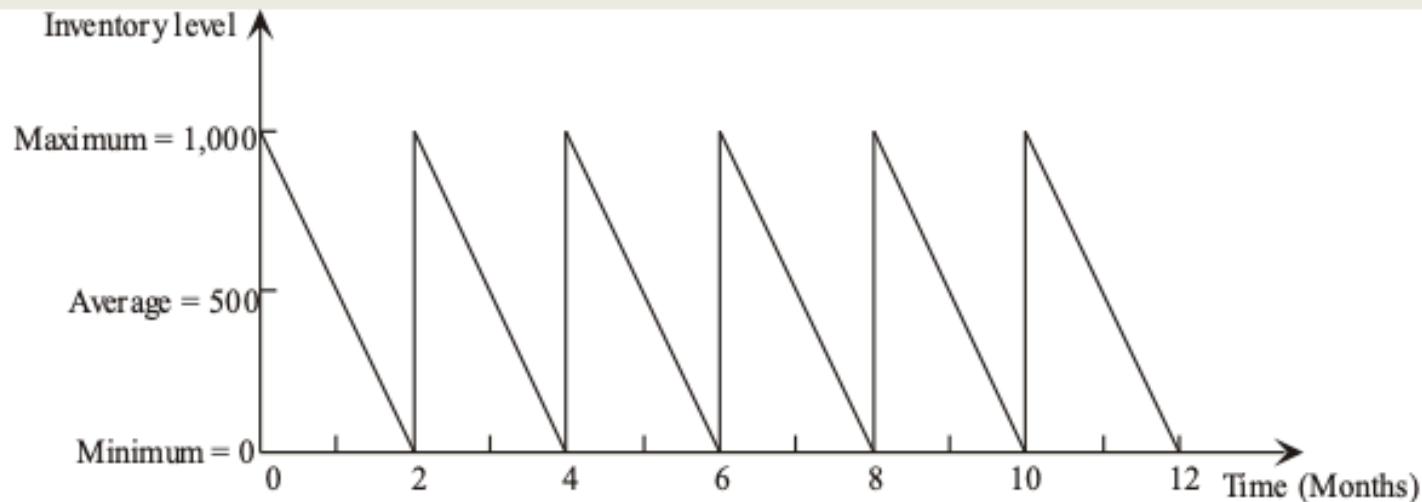
ACT sells tires. When the inventory level of tires gets low, ACT places an order with its supplier to replenish the inventory. Shipments arrive 9 working days after the placement of the order. The following are sales and cost data for ACT:

- 1) The tires sell at a regular rate of about 500 per month (or 16.67 tires per day, assuming 30day/month)
- 2) Policy has been to order 1,000 tires as needed every couple months.
- 3) The order is placed just in time to have the delivery arrive as inventory runs out
- 4) The purchase price is \$20 per tire

ACT Example (cont'd)

- 5) The ordering cost is \$115 per order, including
 - Cost for delivery from the supplier
 - Cost for initiating and processing the purchase order
 - Cost for receiving shipment and placing it into storage
- 6) Cost of holding a tire for a year is \$4.20 or 21% of unit cost including
 - cost of capital tied up in inventory (at 15% of unit cost per year)
 - cost of leasing warehouse space
 - cost of insurance against loss by fire, theft, etc.
 - cost of inventory personnel
 - Holding cost may be expressed as a fixed percentage of unit cost (which may vary – see Discount Model later)
- 7) The annual cost of being out of stock is \$7.50 per tire short, based on these consequences:
 - Customer dissatisfaction
 - Potential price drop to placate a customer due to late deliveries.
 - Delayed revenue
 - The cost of additional record keeping required for out-of-stock tires.

Current Policy: Inventory Levels



Current Policy: Average Cost per Year

Purchase Cost: annual demand = $500 \times 12 = 6000$ units, unit purchase cost = \$20 per tire, annual purchase cost = $6000 \times 20 = \$120,000$

Ordering Cost: number of orders per year = 6, cost per order = \$115, annual ordering cost = $115 \times 6 = \$690$

Holding Cost: average inventory = 500, holding cost per unit = \$4.20 per year, annual holding cost = $500 \times 4.20 = \$2,100$

Shortage Cost: there is no shortage cost under the current policy because no shortage is allowed

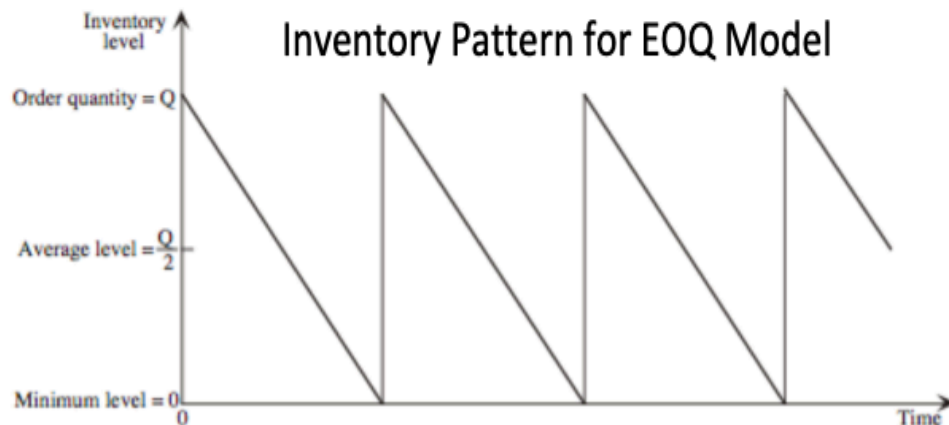
Total annual Cost: $\$120,000 + \$690 + \$2,100 = \$122,790$

The EOQ Model

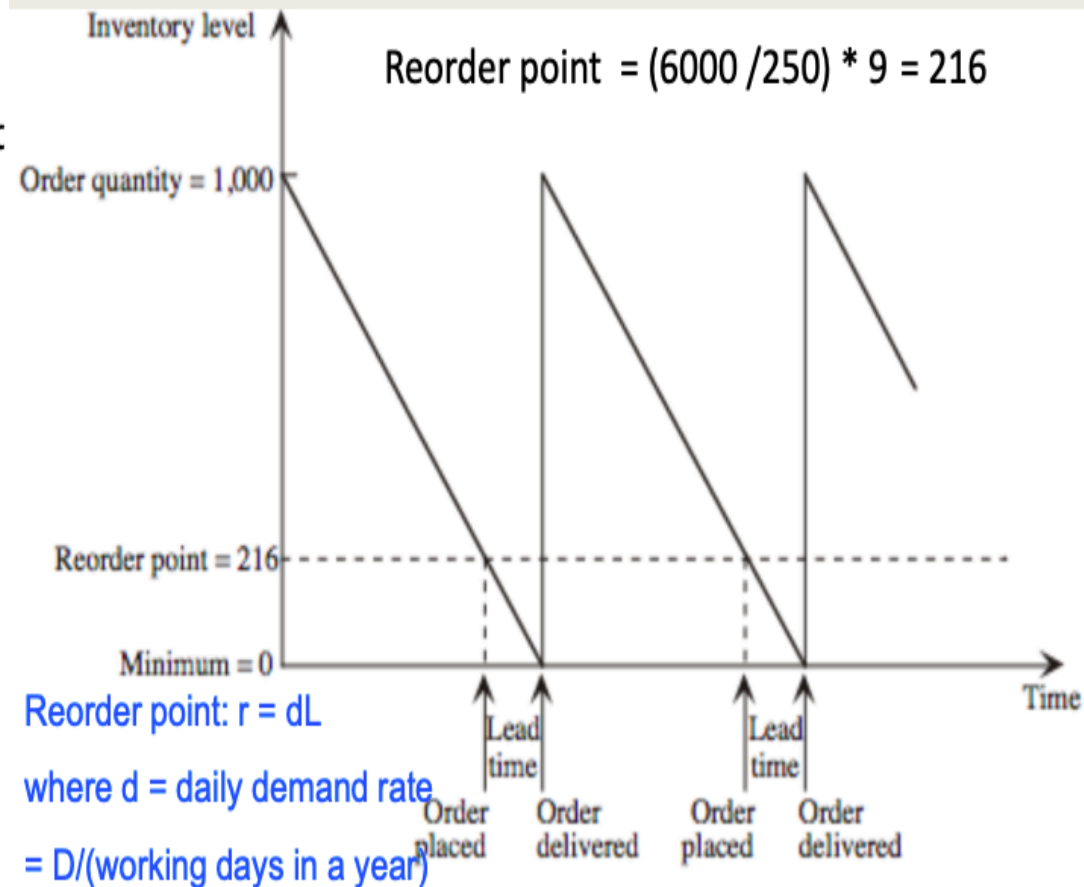
Assumptions:

- A constant demand rate
- The order quantity to replenish inventory arrives all at once just when desired
- No shortages are allowed
- A constant lead time

Let Q = order quantity (the decision to be made).



When to Order: Reorder Point



Total Annual Cost

Let D = annual demand rate
 C = purchase cost per unit
 H = annual holding cost per unit per year
 S = ordering cost per order
 B = shortage cost per unit per year
 TVC = total variable annual cost

1. Annual purchase cost = $C \cdot D$ (constant in this case)
2. Annual ordering cost = $S \cdot \frac{D}{Q}$
3. Annual holding cost = $H \cdot \frac{Q}{2}$
4. Annual shortage cost = 0 (no shortage allowed)
5. $TVC = S \cdot \frac{D}{Q} + H \cdot \frac{Q}{2}$

The EOQ Formula – Minimizing TVC

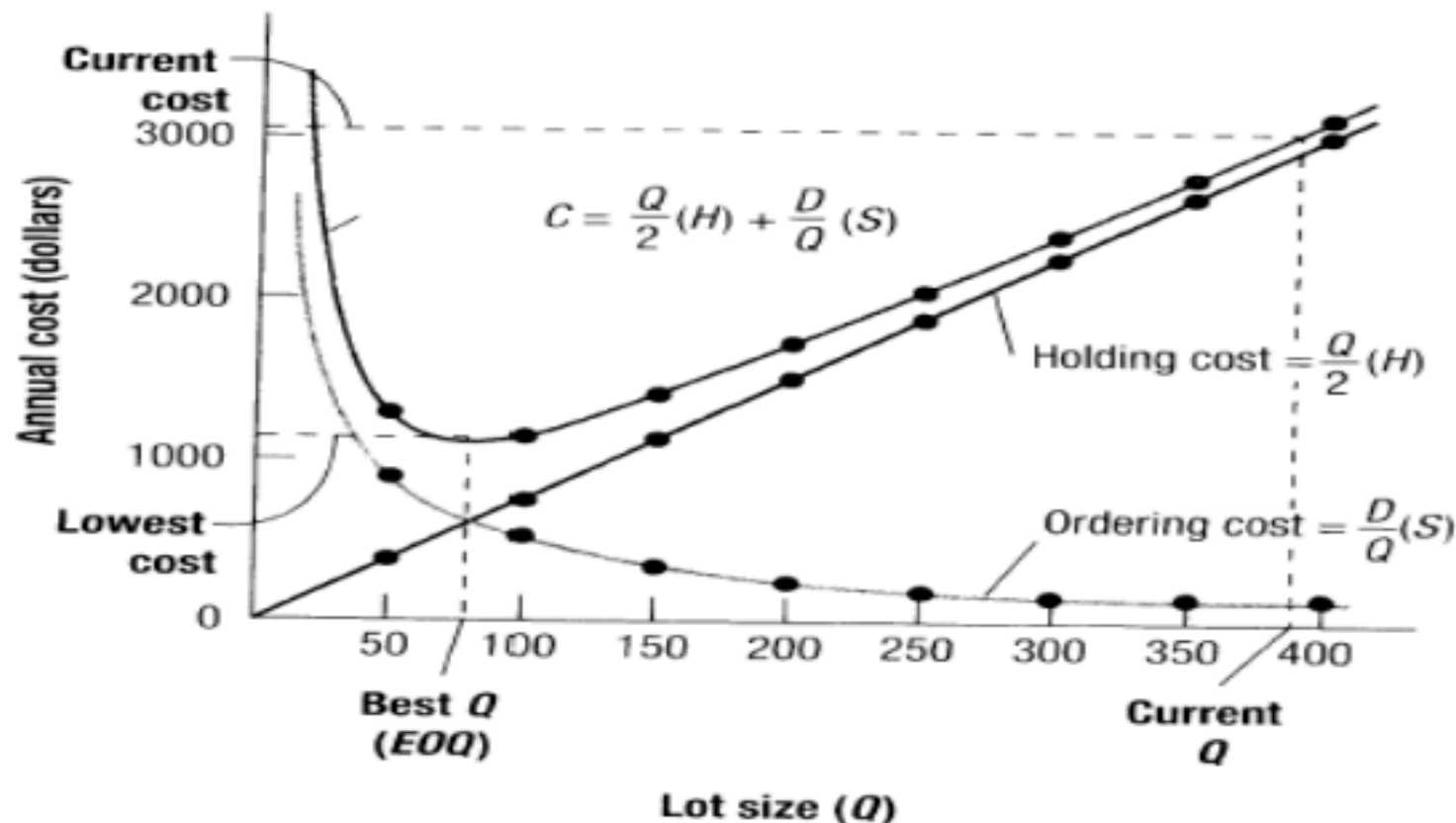
- $TVC = S \frac{D}{Q} + H \frac{Q}{2}$
- The value of Q that minimizes TVC : $Q^* = \sqrt{\frac{2DS}{H}}$
- Number of orders per year: $\frac{D}{Q^*}$
- Time between orders (cycle time): $\frac{Q^*}{D}$ years
- Minimum annual relevant cost (TVC):
$$TVC^* = \left[\frac{1}{2} Q^* H \right] + \left[S \frac{D}{Q^*} \right] = \sqrt{2DSH}$$
- ACT Example: $Q^* = \sqrt{\frac{2 \times 6000 \times 115}{4.2}} = 573$

The Current Policy and the EOQ Policy

Quantity	EOQ Policy	Current Policy
Order quantity	573	1000
Average Inventory level	286.6	500
Annual ordering cost	\$1,203.7	\$690
Annual holding cost	\$1,203.7	\$2,100
Total variable cost	\$2,407.4	\$2,790

1. EOQ achieves a saving of 13.71%
2. Balance of cost components

Insensitivity of EOQ



1. Ordering cost = Ordering cost at Q^*
2. TVC is insensitive to small changes in parameters

Quantity Discounts: the ACT Case

The supplier offers the following quantity discounts, in which a discount is offered depending on the order quantity.

Discount Quantity	Order Quantity	Discount	Unit Cost
1	0 to 749	0	\$20.00
2	750 to 1,999	1%	\$19.80
3	2,000 or more	2%	\$19.60

Inventory Control Policy

What is the optimal ordering quantity?

1. Assume all the conditions for the basic EOQ model
2. Since total purchase cost depends on unit purchase price, which in turn depends on order quantity, we have to include this cost into consideration
3. Let holding cost be a percentage of unit cost or $H = C \cdot I$, where I is the inventory holding cost rate

Suppose the inventory holding cost rate is $I = 21\%$.

Discount Category	Price (C)	Unit Holding Cost $H = I \cdot C = 0.21C$
1	\$20.00	$0.21(\$20) = \4.20
2	\$19.80	$0.21(\$19.80) = \4.158
3	\$19.60	$0.21(\$19.60) = \4.116

Inventory Control Policy (cont'd)

4. The best order quantity for a specific price is the EOQ or the minimum qualifying quantity, whichever is larger

$$TVC = CD + S \frac{D}{Q} + H \frac{Q}{2}$$

For the ACT example, consider the unit price of \$19.60 (the lowest)

$$EOQ = \sqrt{(2 * 115 * \frac{6000}{4.116})} = 579$$

Best order quantity to qualify for this price: 2000

$$\begin{aligned} TVC &= 6000 * 19.6 + 115 * 6000 / 2000 + 4.116 * 1000 \\ &= 122,061 \end{aligned}$$

Inventory Control Policy (cont'd)

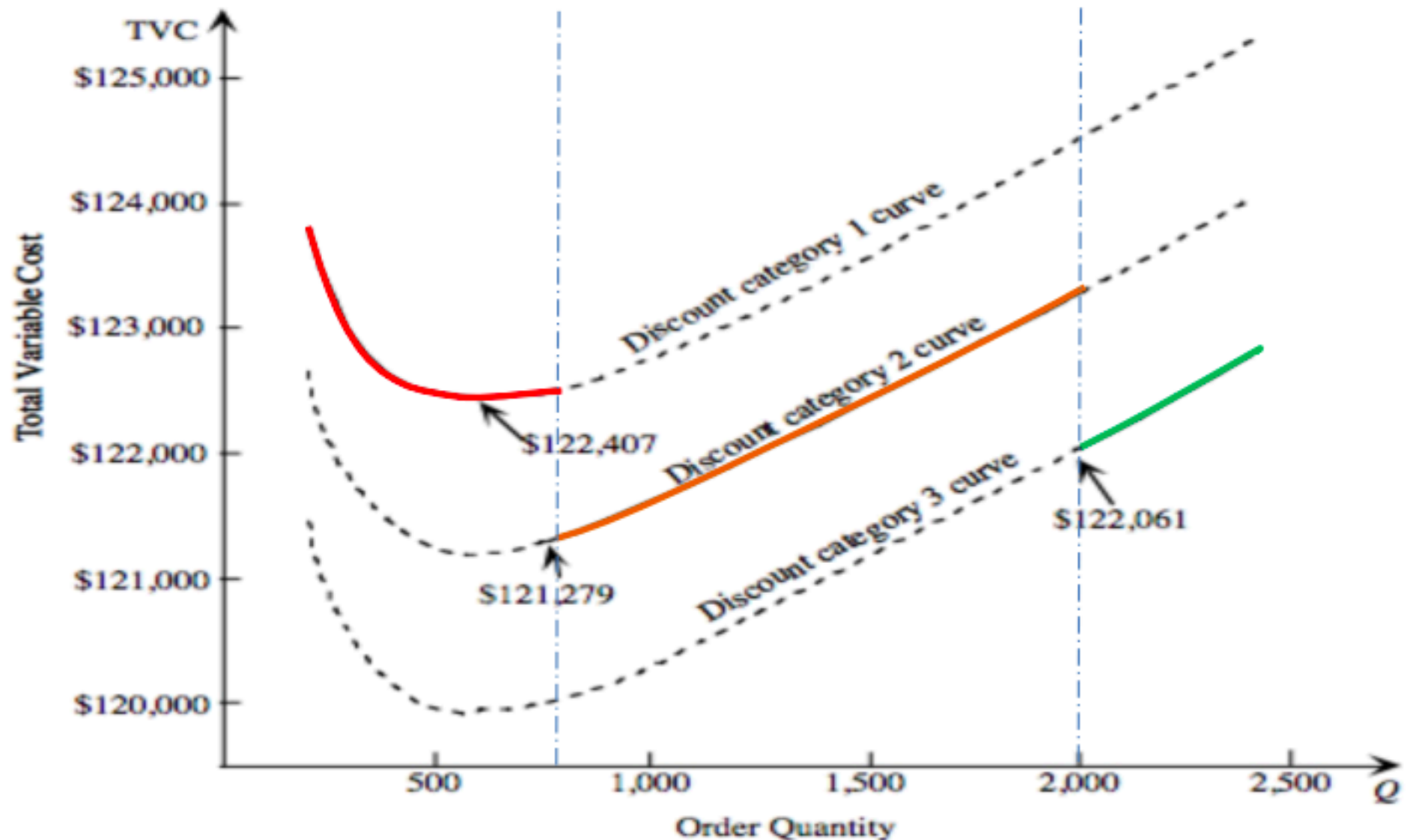
5. Repeat Step 4 for each price category and find the best order quantity and total cost

Discount Category	Order Quantity	Annual Costs			Total
		Purchase	Ordering	Holding	
1	573	\$120,000	\$1,204	\$1,204	\$122,407
2	750	\$118,800	\$920	\$1,559	\$121,279
3	2,000	\$117,600	\$345	\$4,116	\$122,061

Evaluate EOQ for all categories. If EOQ is within category's range, then that's the lowest cost order quantity.

Otherwise use the range limit closer to the EOQ as order quantity.

Total Variable Cost vs Order Quantity

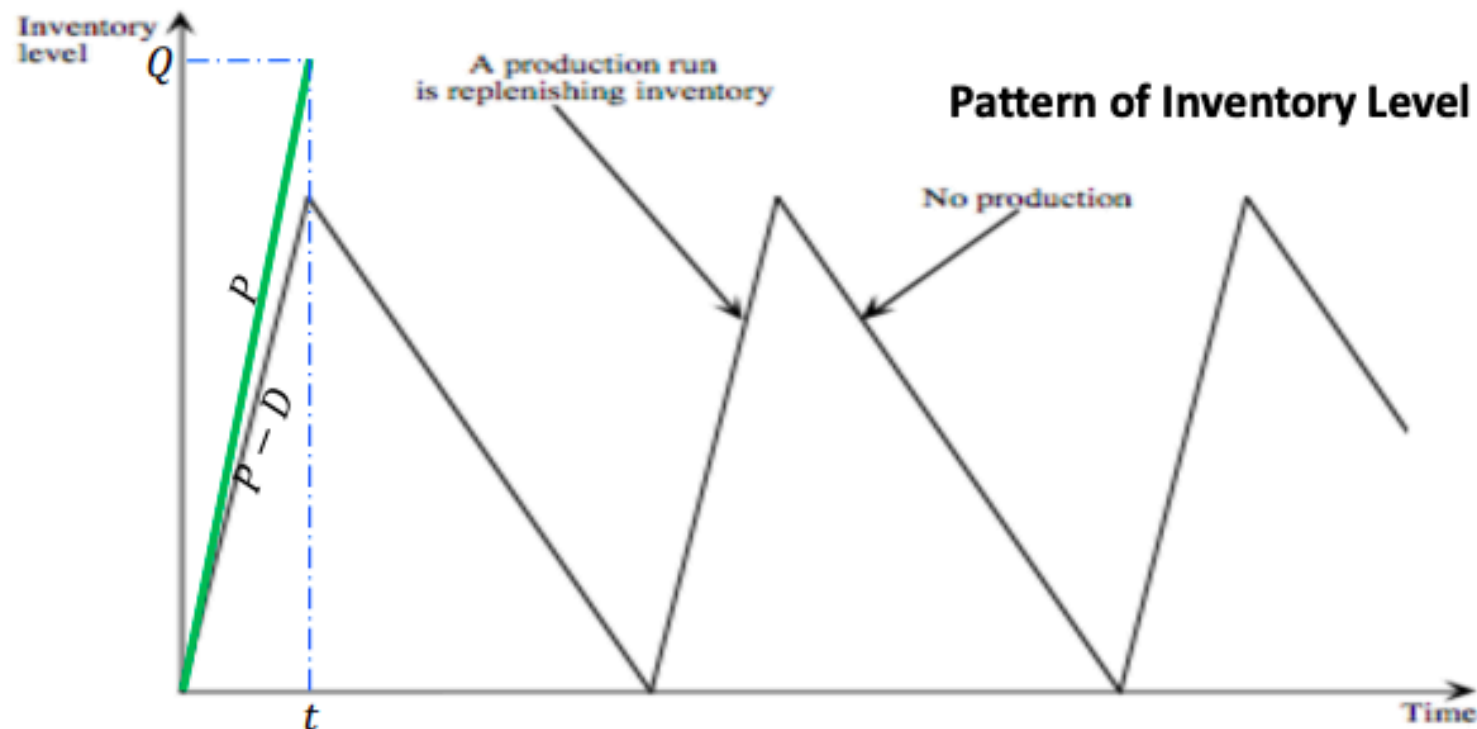


Production Batch Size

Assumptions:

- The item is produced internally at a constant rate of P ($P > D$)
- Assume the conditions for the basic EOQ model (i.e. no shortage, etc.)
- A production run is scheduled to begin each time when the inventory level drops to 0, and this production replenishes inventory at a constant rate throughout the duration of the run.

Decision variable: Q = production batch size



Optimal Production Batch Size

$$\text{Number of production runs per year} = \frac{D}{Q}$$

$$\text{Production rate (per year)} = P$$

$$\text{Maximum inventory level} = Q \left(1 - \frac{D}{P} \right)$$

$$\text{Average inventory level} = \frac{Q}{2} \left(1 - \frac{D}{P} \right)$$

$$TVC = H \frac{Q}{2} \left(1 - \frac{D}{P} \right) + S \frac{D}{Q} \quad (\text{holding} + \text{ordering})$$

$$\text{Optimal production lot-size: } Q^* = \sqrt{\frac{2DS}{H \left(1 - \frac{D}{P} \right)}}$$

$$\text{Minimum annual cost: } TVC^* = \sqrt{2DSH \left(1 - \frac{D}{P} \right)}$$

An Example: the SOCA Problem

SOCA, a television manufacturing company, produces its own speakers for assembly into television sets. Current policy for managing SOCA's inventory of speakers:

- Daily demand rate = 1,000 speakers per day.
- Daily production rate = 3,000 speakers per day (when producing).
- The production facilities get set up to start a production run each time when the inventory level is scheduled to drop to 0.
- Each production run produces 30,000 speakers over a period of 10 working days, so another 20 working days elapse before the next production run is started.
- Annual holding cost is \$3.60 per speaker and setup cost is \$12,000 each time

Cost Analysis for the Current Policy (1)

Assume 250 working days per year

Max inv. level = Q minus demand during production run
= $30,000 - (10 \text{ days})(1,000 \text{ speakers / day})$
= 20,000 speakers (or $30000(1-1000/3000)$)

Average inv. level = $(1/2)$ (max inv. level) = 10,000 speakers

Annual holding cost = $10,000 * 3.60 = \$36,000$

Annual setup cost = $12000 * 250 * 1000 / 30000 = \$100,000$

TVC = \$136,000 -- is this optimal?

Economic Production Batch Size

Optimal Batch Size and minimum cost:

$$Q^* = \sqrt{\frac{2(12,000)(250,000)}{3.60\left(1 - \frac{250,000}{750,000}\right)}} = 50,000,$$

$$TVC = \$12,000 \frac{250,000}{50,000} + \$3.60(25,000)\left(1 - \frac{250,000}{750,000}\right) = \$120,000.$$

Inventory Model Rationale

A cost reduction of 11.76%.

Large inventory:

- Economy of scale
- Smooth operation
- Good customer service



Small Inventory :

- Low inventory carrying costs
- Low scrap cost
- Increased productivity and quality
- Flexibility for technological changes

[illegible]

Review of the EOQ Analysis

- Demand for the product is constant and uniform throughout the period
- All demands for the product will be satisfied (No backorders are allowed)
- Lead time (from ordering to receipt) is constant
- Price per unit of product is constant
- Inventory holding cost is based on average inventory
- Ordering or setup costs are constant

An Approximate Determination

Determination of the order quantity Q:

Use the deterministic EOQ analysis (basic EOQ or EOQ with planned shortage)

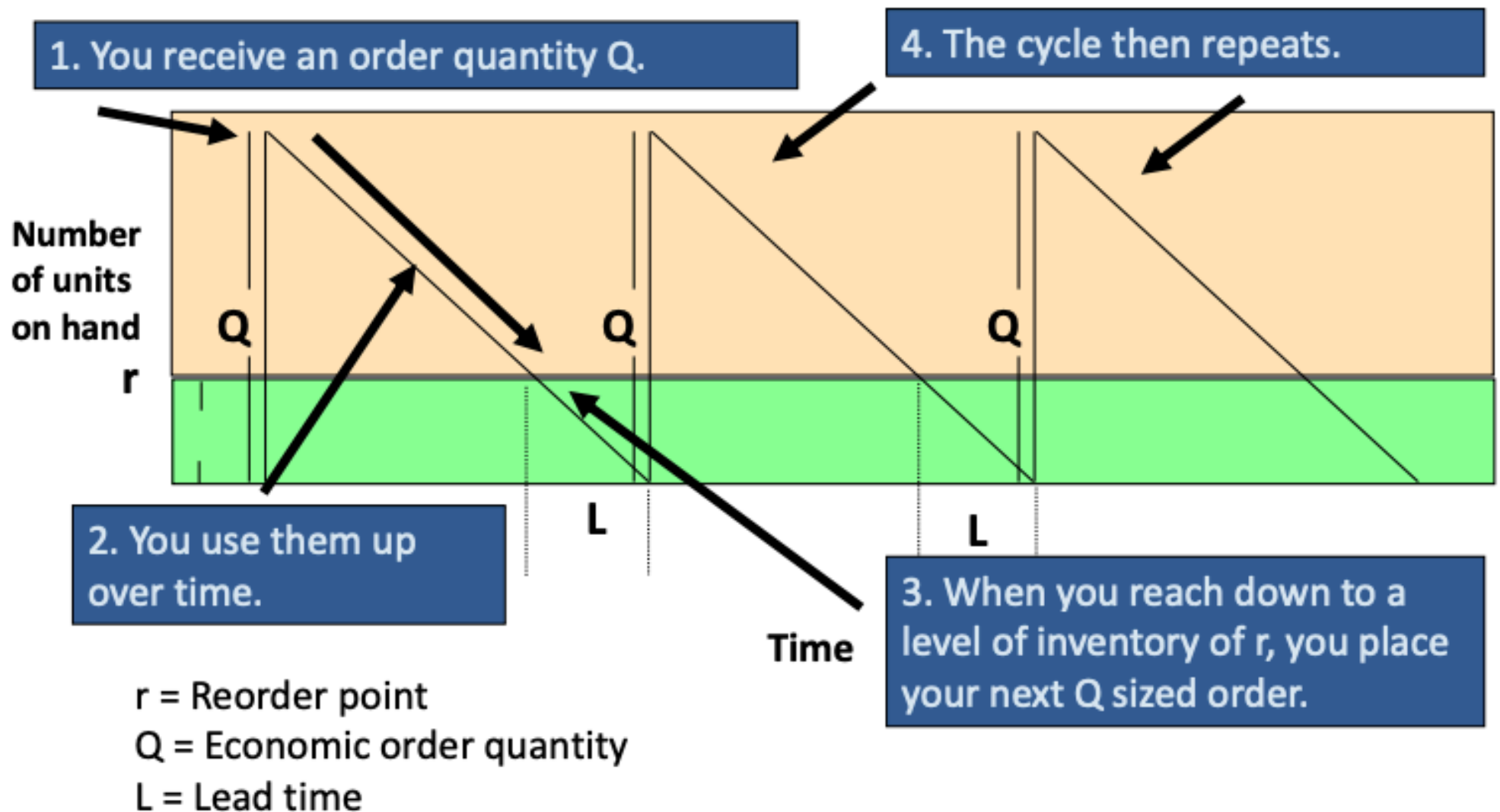
Determination of the re-order point r:

Set a service level $1 - \alpha$ so that the chance of stockout is equal to α or

$$P(d_L \leq r) = 1 - \alpha$$

where d_L is the lead time demand (a random variable)

Behavior of the Basic EOQ Model



An Example: Robert's Drug

Robert's Drugs is a drug wholesaler supplying 55 independent drug stores. Roberts wishes to determine an optimal inventory policy for the *Comfort* brand headache remedy. Sales of *Comfort* are relatively constant as the past 10 weeks of data indicate

<u>Week</u>	<u>Sales (cases)</u>	<u>Week</u>	<u>Sales (cases)</u>
1	110	6	120
2	115	7	130
3	125	8	115
4	120	9	110
5	125	10	130

An Example: Robert's Drug (cont'd)

Each case of *Comfort* costs Roberts \$10 and Roberts uses a 14% annual holding cost rate for its inventory. The cost to prepare a purchase order for *Comfort* is \$12.

The lead time for a delivery of *Comfort* has averaged four working days. Lead time demand has therefore been estimated as having a normal distribution with a mean of 80 cases and a standard deviation of 10 cases. Roberts wants at most a 2% probability of selling out of *Comfort* during this lead time.

What order quantity and reorder point should Roberts use to control the inventory of *Comfort*?

Determination of Order Quantity

Demand:

Weekly average = 120 (cases) (calculate from the data)

Average annual demand rate $D = 120 * 52 = 6240$ (cases)

Ordering cost:

$S = \$12$ per order

Holding cost:

Unit purchase price = \$10,

Annual inventory holding cost rate = 14%

$H = 10 * 0.14 = \$1.4$ per unit per year

$Q^* = 2 * (6240) * 12 / 1.4 = 327.0649$

Determination of Re-order Point

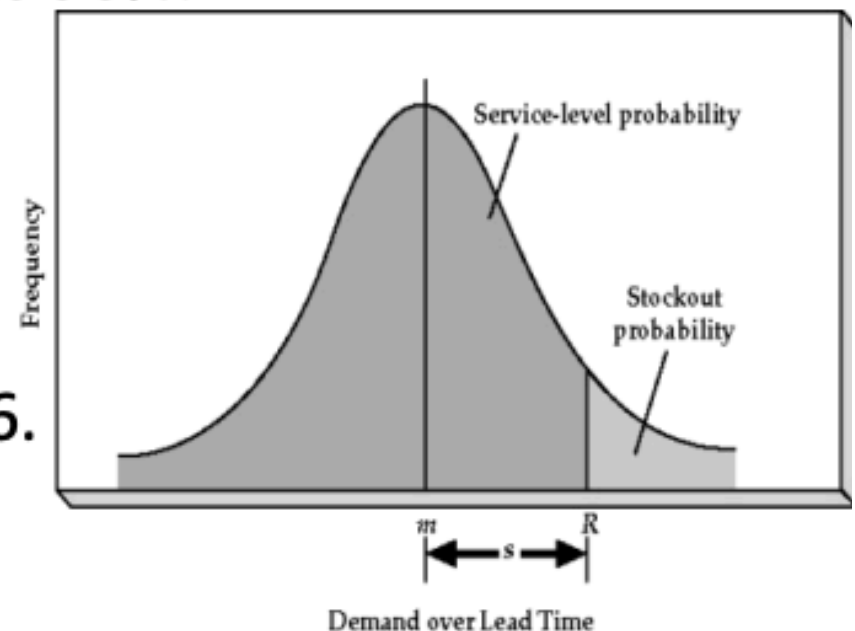
Lead time demand is normally distributed with $\mu = 80$, $\sigma = 10$. Since Roberts wants at most a 2% probability of selling out of *Comfort*, the service level is 98%.

Reorder point: $r = \mu + z\sigma$

Safety stock: $z\sigma$

The corresponding z value is 2.06.

That is, $P(z > 2.06) = 0.0197$.



Hence Roberts should reorder *Comfort* when supply reaches $r = \mu + z\sigma = 80 + 2.06(10) = 100.6$ cases. The safety stock is $z\sigma = 20.6$ cases.

Estimation of System Cost

$$\begin{aligned}\text{Average inventory} &= \text{average normal stock} + \text{safety stock} \\ &= \frac{1}{2}(Q) + \text{safety stock } (z\sigma)\end{aligned}$$

$$\begin{aligned}\text{Total annual cost} &= \text{Ordering cost} + \text{Cost of holding normal stock} + \text{Cost of holding safety stock} \\ &= S \frac{D}{Q^*} + H \frac{Q^*}{2} + z\sigma H\end{aligned}$$

Inventory related cost for Roberts

$$\text{Ordering: } S \frac{D}{Q^*} = ((6240)(12)/327) = \$229$$

$$\text{Holding-Normal: } H \frac{Q^*}{2} = (1/2)(327)(1.40) = \$229$$

$$\text{Holding-Safety Stock: } z\sigma H = (1.40)(21) = \$29$$

$$\text{Total} = \$487$$

A Periodic-Review System

- In practice, inventory is often checked and replenished only at specified points in time, e.g. at the end of each week, month, etc.
- A Periodic-Review Order-up-to Policy
 - Inventory position is reviewed in a constant interval of T time units
 - Ordering decisions are made only at points of review
 - Replenishment policy: order the needed stock at a review point to raise the inventory position to a fixed replenishment level M .

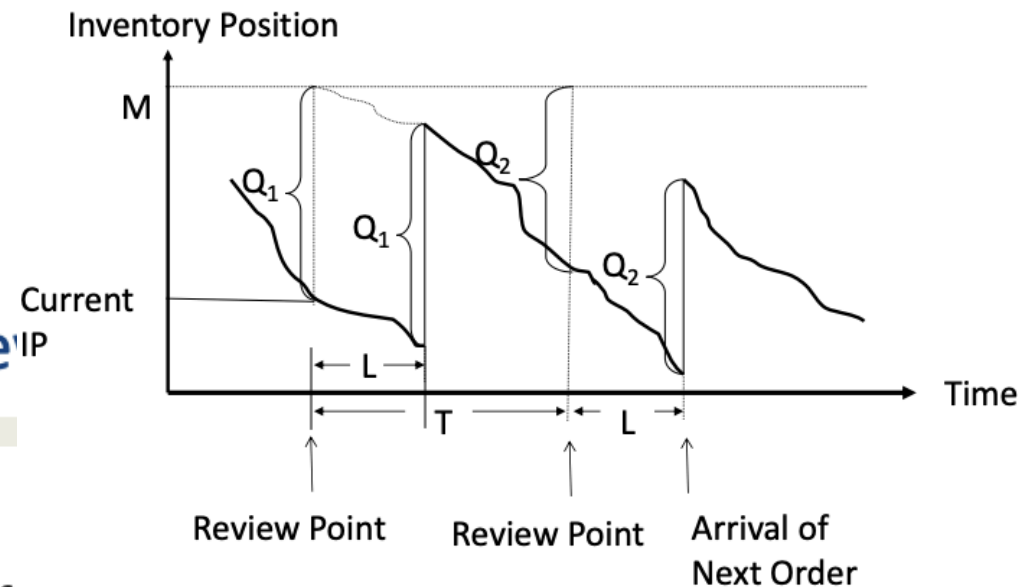
Continuous-review & Periodic-Review

Continuous-Review

- Monitor stock in real time
- Use and provide updated demand/stock information
- Implementation requires an information system
- The continuous-review (r, Q) policy is a stock-based order-up-to policy: let $M = r + Q$ and order the needed stock to raise the inventory position to M whenever the inventory position drops to r .

Periodic-Review

- Review and order inventory in a fixed interval of length T time units
- Do not use updated information in decision making
- Easy to implement



Determination of M

The stock M is to meet the demand in an interval of length $L + T$, which is also known as the protection interval.

Let d_{L+T} denote the (random) demand in this period

Define service level in terms of the probability of no stockout during this period to be $1 - \alpha$

Determination of M: find the minimum M to meet the service level $1 - \alpha$ or $P(d_{L+T} \leq M) \geq 1 - \alpha$

When demand follows a normal distribution:

$$M = \mu_{L+T} + z \sigma_{L+T}$$

An Example: Ace Brush

Joe Walsh is a salesman for the Ace Brush Company. Every three weeks he contacts Dollar Department Store so that they may place an order to replenish their stock. Weekly demand for Ace brushes at Dollar approximately follows a normal distribution with a mean of 60 brushes and a standard deviation of 9 brushes.

Once Joe submits an order, the lead time until Dollar receives the brushes is one week. Dollar would like at most a 2% chance of running out of stock during any replenishment period. If Dollar has 75 brushes in stock when Joe contacts them, how many should they order?

If the annual holding rate per unit is 15% of unit cost, each brush costs \$5 and each order costs \$10, calculate annual total inventory cost. Assume each year has 40 working weeks.

An Example: Ace Brush (cont'd)

The review period plus the lead time totals 4 weeks.

Weekly demand

Distribution:

Normal

Mean:

60

Standard deviation:

9

Review Period plus Lead Time

Distribution:

Normal

Mean:

$4 \times 60 = 240$

Standard deviation:

$2 \times 9 = 18$

Replenishment level:

$M = 240 + 2.05(18) = 276.9$

Order Quantity:

$Q = 276.9 - 75 = 201.9$

Total Annual Inventory Cost

Total Annual Inventory Cost = Annual cycle stock holding cost + Annual safety stock holding cost + Annual ordering cost

Let $\bar{d}$ be mean daily demand.

$$\text{Annual cycle stock holding cost} = \frac{\bar{d}T}{2} H$$

$$\text{Annual safety stock holding cost} = z\sigma_{L+T} H$$

$$\text{Annual ordering cost} = \frac{D}{\bar{d}T} S$$

$$\text{Total Annual Inventory Cost} = \frac{\bar{d}T}{2} H + z\sigma_{L+T} H + \frac{D}{\bar{d}T} S$$

An Example: Ace Brush (cont'd)

$$I = 15\%, C = \$5, H = IC = 0.15 \times 5 = 0.75$$

$$\text{Mean weekly demand } \bar{d} = 60$$

$$\text{Annual demand } D = \bar{d} \times 40 = 2,400$$

$$\text{Annual cycle stock holding cost} = \frac{\bar{d}T}{2} H = \frac{60 \times 3}{2} \times 0.75 = 67.5$$

$$\text{Annual safety stock holding cost} = z\sigma_{L+T}H = 2.05 \times 18 \times 0.75 = 27.675$$

$$\text{Annual ordering cost} = \frac{D}{\bar{d}T} S = \frac{2400}{60 \times 3} \times 10 = 133.333$$

$$\text{Total Annual Inventory Cost} = 67.5 + 27.675 + 133.333 = \$ 228.51$$

Decision Criteria

Criteria

- To maximize the maximum – optimistic
- To maximize the minimum -- pessimistic,
- To minimize regret,
- To maximize/minimize the expected value, etc.

Decision context

- Decision situation --- the context affects the objectives and criteria
- Decision maker

Optimistic and Pessimistic

Profit in \$millions

	S1	S2	S3
D1	0.5	0.4	0.3
D2	1.0	0.6	0.1
D3	2.0	0.5	-0.5

Profit in \$millions

	S1	S2	S3	Max	Min
D1	0.5	0.4	0.3	0.5	0.3
D2	1.0	0.6	0.1	1.0	0.1
D3	2.0	0.5	-0.5	2.0	-0.5

Maximin
Decision

Maximin
Payoff

Maximax
Decision

Maximax
Payoff

Regret Table

	S1	S2	S3	Max
D1	1.5	0.2	0.0	1.5
D2	1.0	0.0	0.2	1.0
D3	0.0	0.1	0.8	0.8

Minimax
Regret
Decision

Minimax
Regret

Calculating Expected Values with a Payoff Table

Profit in \$millions

	S1	S2	S3	EV
D1	0.5	0.4	0.3	0.38
D2	1.0	0.6	0.1	0.49
D3	2.0	0.5	-0.5	0.35
Prob.	0.1	0.6	0.3	

Decision Tree

- A graphical representation of a decision problem according to the logical sequence in their time of occurrence for events
- Consists of Decision, Event, and Payoff nodes
- Drawn in a left-to-right manner indicating events unfolding from early to later stages.
- Able to display multi-stage decision problem by listing exhaustively all possible decision and event paths.

Notation and Symbol

Decision event: square



Chance (uncertain) event: circle



Value (payoff): rounded rectangle

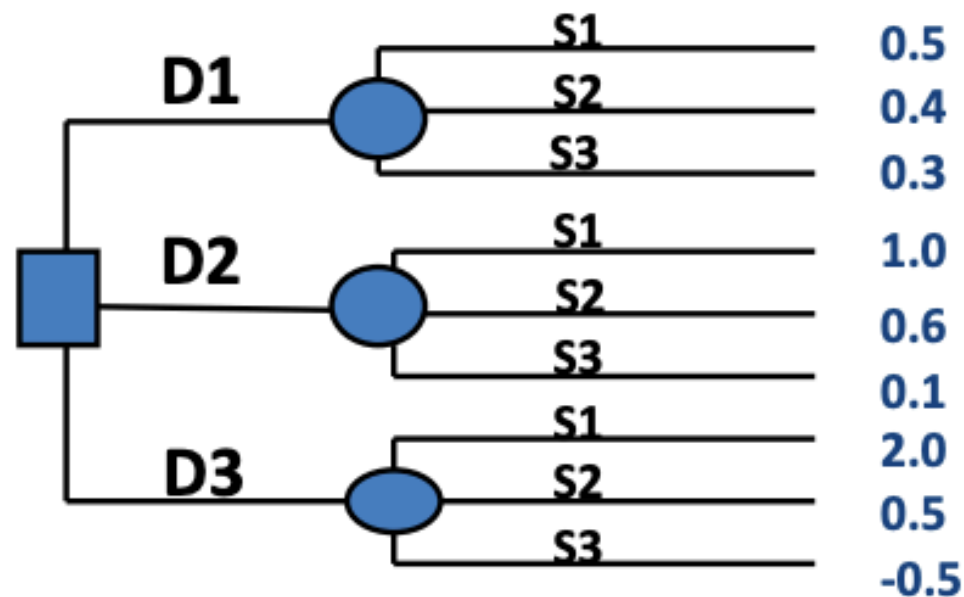


Link: pointed arrow or arc



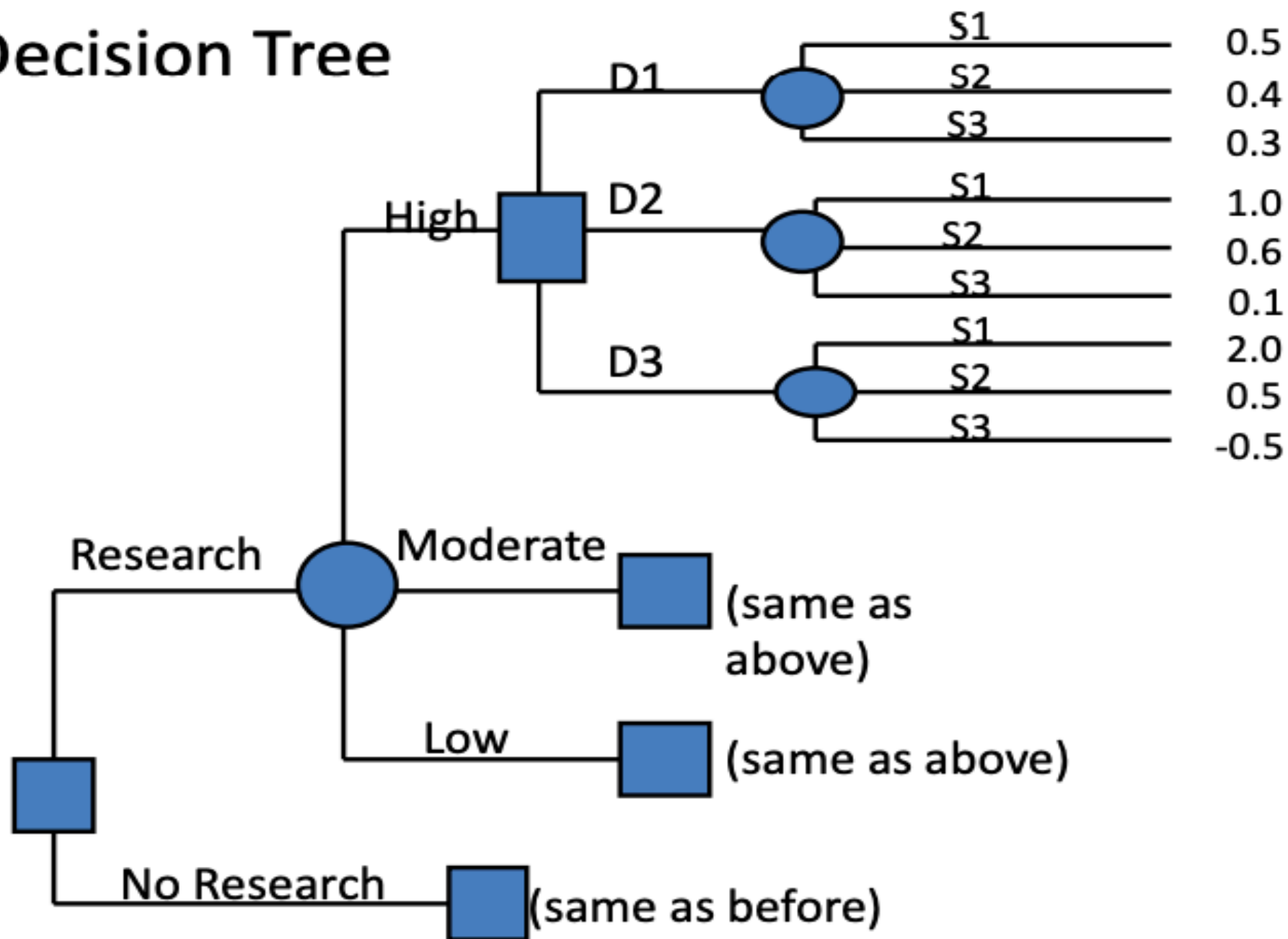
- Use a square for a decision event and a circle for a chance event
- Use a link leaving a chance event for a different outcome (state of nature), and a link leaving a decision node for a different decision alternative.
- Display the payoff at the end of each limb of a tree (the result for the series of branches that make up that limb).

Decision Tree: the Marketing Example



- To calculate correctly the expected value in order to make a correct decision at the root node, the payoff nodes should have net payoff after adding any bonus payoff and subtracting any cost payoff along the path.

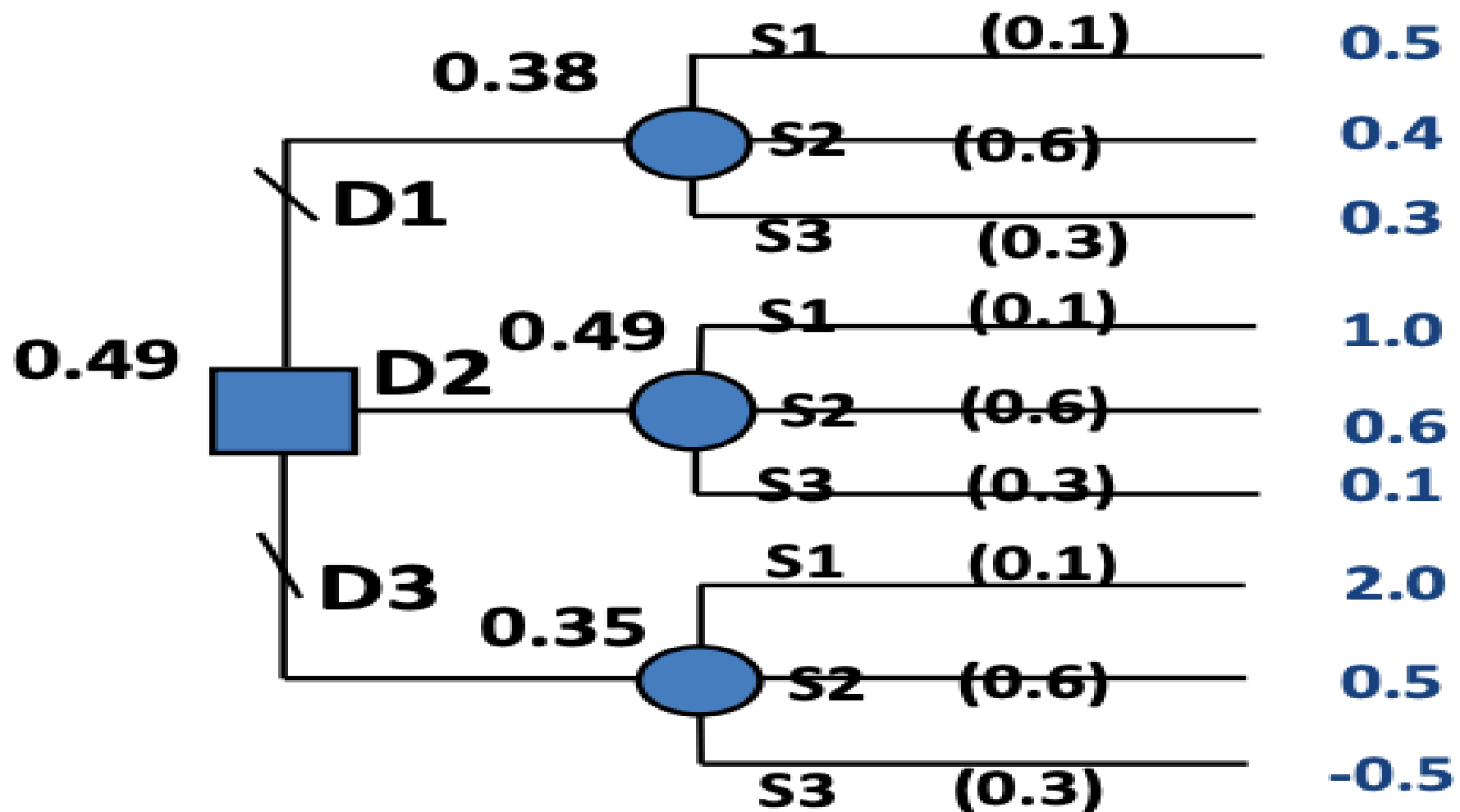
Decision Tree



Evaluating A Decision Tree

- Start from the “end branches” and work leftwards towards the “root”
 - For each chance node, calculate the EV and write the EV above the node
 - For each decision node, select and mark the decision with the maximum EV. Write the optimal EV above the decision node
- The optimal sequence of decisions is given by the route marked. The optimal EV is given above the “root” node

Marketing Example: Basic Case



The Bayes Rule

1. Prior belief
2. New information such as forecast, new research findings, experts' opinions, etc.
3. The prior belief is revised in the light of the new information
4. The updating depend on the prior probabilities and the reliability of the new information source

For the Marketing Example ...

1. Prior belief: $P(S1) = 0.1$, $P(S2) = 0.6$, $P(S3) = 0.3$
2. New forecast is high (denoted by H)
3. How reliable is the market research company?
4. How to revise the probabilities?

The Bayes Theorem

- S_j state of nature $j = 1, \dots, n$
- I_k market research indicator $k = 1, \dots, L$
- $P(I_k | S_j)$: reliability of the marketing research

$$P(S_j | I_k) = \frac{P(I_k | S_j)P(S_j)}{\sum_{t=1}^n P(I_k | S_t)P(S_t)}$$

Tableau method

- I_k , $k = 1, 2, 3$ the event that the market research company forecasts high, moderate, and low market acceptance level
- From past performance, the reliability of the market research company is given in the following table.
- Reliability is the conditional probability of the column event given the row event

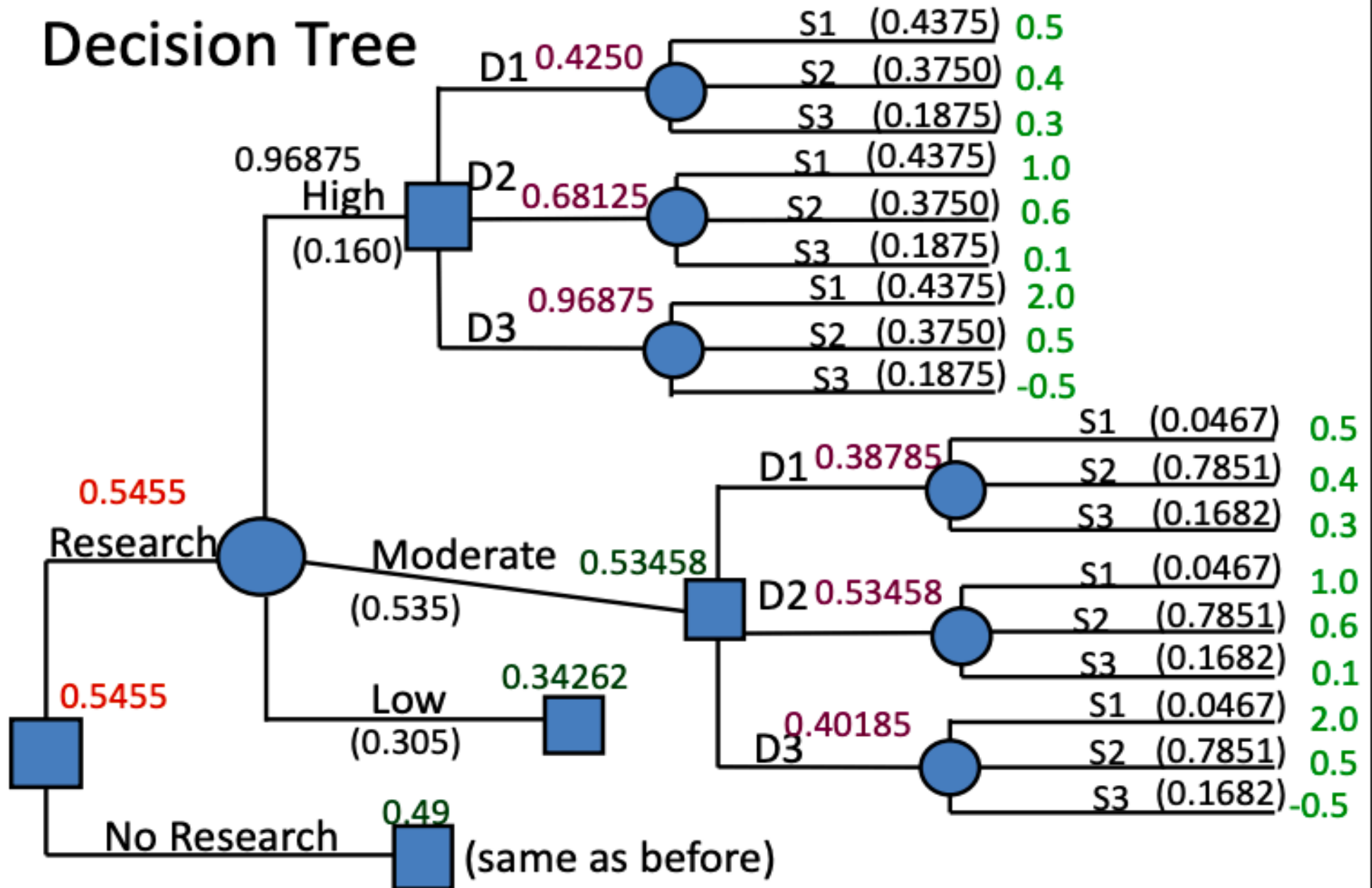
Reliability of the market research

	$P(I_k \mid S_j)$		
	I_1	I_2	I_3
S_1	0.70	0.25	0.05
S_2	0.10	0.70	0.20
S_3	0.10	0.30	0.60

Tableau

	$P(S_j)$	$P(I_k S_j)$			$P(I_k \cdot S_j)$			$P(S_j I_k)$		
		I_1	I_2	I_3	I_1	I_2	I_3	I_1	I_2	I_3
S_1	0.1	0.70	0.25	0.05	0.07	0.025	0.005	0.4375	0.0467	0.0164
S_2	0.6	0.10	0.70	0.20	0.06	0.420	0.120	0.3750	0.7851	0.3934
S_3	0.3	0.10	0.30	0.60	0.03	0.090	0.180	0.1875	0.1682	0.5902
		$P(I_k)$			0.16	0.535	0.305			

Decision Tree



(same as above)

The Case of Perfect Information

Outcomes:

S1: High

S2: Moderate

S3: Low

D1: Singapore

D2: Asia

D3: International

Payoffs

	S1	S2	S3
D1	0.5	0.4	<u>0.3</u>

D2	1.0	<u>0.6</u>	0.1
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D3	<u>2.0</u>	0.5	-0.5
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Prob.	0.1	0.6	0.3
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	0.2	0.36	0.09
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EVwPI = 0.65

EVPI & EVSI

Expected Value of Perfect Information

- Expected Value with Perfect Information or
 $EVwPI = 2 \times 0.1 + 0.6 \times 0.6 + 0.3 \times 0.3 = 0.65$
- Expected Value of Perfect Information or
 $EVPI = 0.65 - 0.49 = 0.16$

Had you known in advance that S1 is going to happen, what action will you take?

Expected Value of Sampled Information

- Expected value with sample information:
 $EVwSI = 0.5455$ (from the decision tree)
- Expected value of sample information:
 $EVSI = 0.5455 - 0.49 = \$0.0555 \text{ million} = \$55,000$
- Efficiency of Information
 $EVSI/EVPI = 0.0555/0.16 = 34.69\%$

Optimal strategy

- If the fee to be charged is greater than \$55,000 then no survey should be implemented and the optimal strategy is D_2 : market the product in Asia only
- If the fee to be charged is less than \$55,000 then the optimal strategy is to implement the survey. The optimal strategy is then dependent upon the sample outcome

What you have to write as answer

Survey outcome

- If I_1 : then D_3 market the product internationally
- If I_2 : then D_2 market the product in Asia
- If I_3 : then D_1 market the product in Singapore

	A	B	C	D	E	F	G	H	I	J
1		State of	Nature							
2		High Demand	Medium Demand	Low Demand						
3	Decision Alternative	s1	s2	s3						
4	Order 1 lot,	60	60	50						
5	Order 2 lots	80	80	30						
6	Order 3 lots	100	70	10						
7										
8										
9	Decision Alternative	Prior	D1	D2	D3			Max		
10	High Demand	s1	0.3	60	80	100		100	For EVwPI calculations	
11	Medium Demand	s2	0.3	60	80	70		80		
12	Low Demand	s3	0.4	50	30	10		50		
13										
14		EV		56	60	55		60	<-- Max EV	
15								74	<-- EVwPI	

[illegible]